

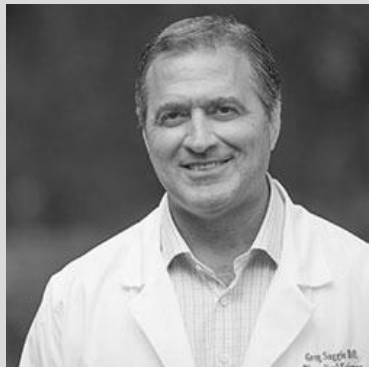
Laboratory Medicine & Anemias

Part 2: Normocytic & Macrocytic Anemia

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I am available to address any questions/comments/concerns
and provide additional resources or discussion.

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Learning Objectives

1. Identify the most common causes of microcytic, normocytic, and macrocytic anemia.
2. Describe the pathophysiology, clinical symptomatology, and laboratory findings of the various causes of microcytic, normocytic, and macrocytic anemia.
3. Illustrate the biochemical basis of heme synthesis, vitamin B12 and folate deficiency, and the clinical approach to such patients.

Lecture Layout

- General Concepts of Anemia
- Microcytic Anemia
- **Normocytic Anemia**
- **Macrocytic Anemia**

Macrocytic Anemias

(MCV $>$ 100)

Macrocytic Anemias

- **Megaloblastic Anemias**: impaired DNA synthesis and **hypersegmented neutrophils**
 - **Folate (B₉ Deficiency)**
 - **Cobalamin (B₁₂ Deficiency)**
 - **Orotic aciduria (*less common*)**
 - **Medications** (hydroxyurea, phenytoin, methotrexate, sulfa drugs)
- **Non-Megaloblastic Anemia**: DNA synthesis is normal, no hypersegmented neutrophils
 - **Chronic alcohol overuse**
 - **Liver disease**
 - **Diamond-Blackfan anemia**
 - **Drugs (5-fluorouracil)**



Megaloblastic Anemias

Megaloblastic Anemia: impairment of DNA synthesis leads to ineffective hematopoiesis

- Abnormally large erythroid precursors and red cells

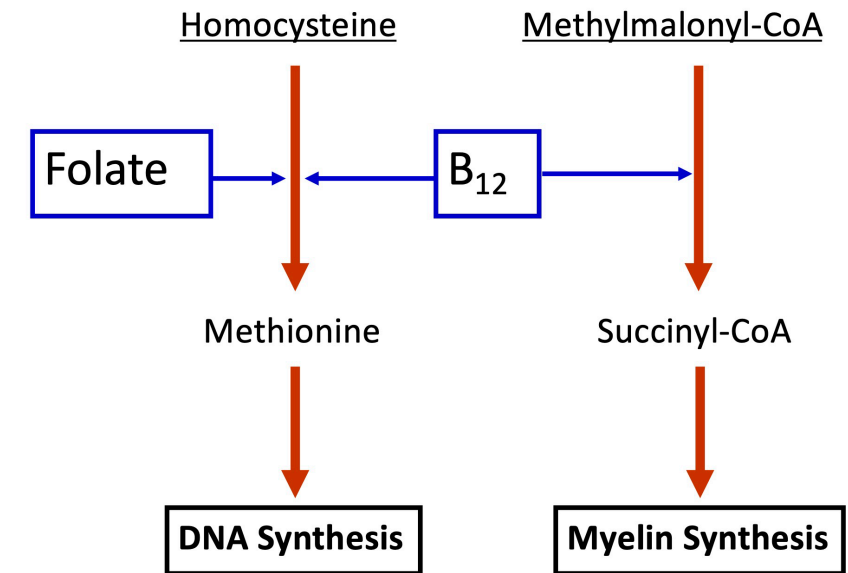
Biochemistry:

- Both folate and Vit B₁₂ are essential for the synthesis of DNA precursors

Simplified Pathway:

- Folate is the main vitamin associated with single carbon transport
- Circulates as methyltetrahydro**folate** (methyl THF)
- Vitamin B₁₂ takes the methyl group off methyl-THF to help convert **homocysteine to methionine** via the enzyme methionine synthase
- Additionally, Vitamin B₁₂ is used in fatty acid metabolism, converting methylmalonyl-CoA to Succinyl-CoA
 - Succinyl-CoA is used as intermediary in the TCA cycle
 - Succinyl-CoA is also used in **myelin synthesis**

Metabolic Pathways for Folate and Vitamin B₁₂



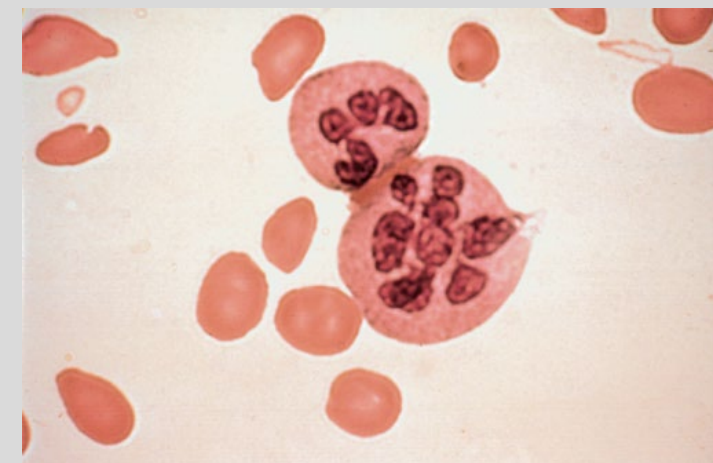
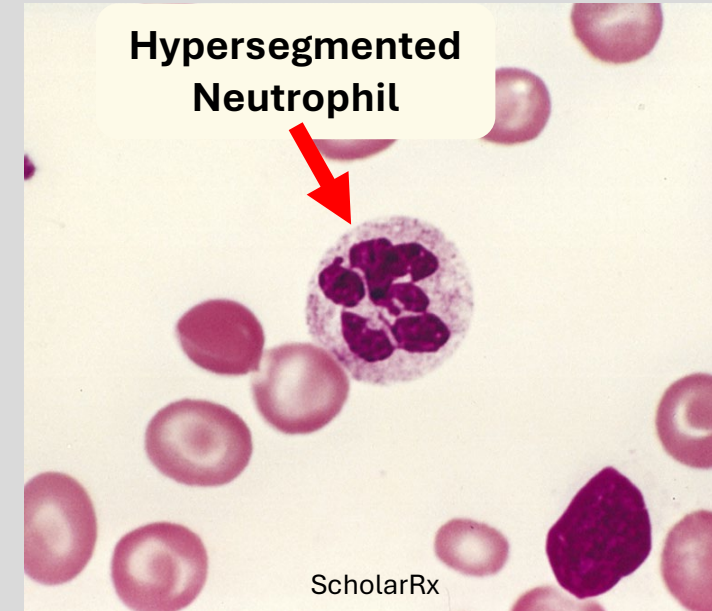


Megaloblastic Anemias

Biochemistry:

- Both folate and Vit B₁₂ are essential for the synthesis of DNA precursors
- Without these DNA precursors, the nuclei of rapidly dividing cells remain larger
- Immature nuclei do not have a lot of DNA; as DNA is made, nuclei becomes smaller and more condensed
 - Impaired division and enlargement in RBCs: megaloblastic anemia with LARGE nuclei
 - Impaired division and enlargement in granulocytic precursors leads to **hypersegmented neutrophils**
 - *These changes can be seen in any rapidly dividing cell, will see a larger nucleus*
- Connection: homocysteine increases risk to thrombosis via endothelial damage (including myocardial infarctions)

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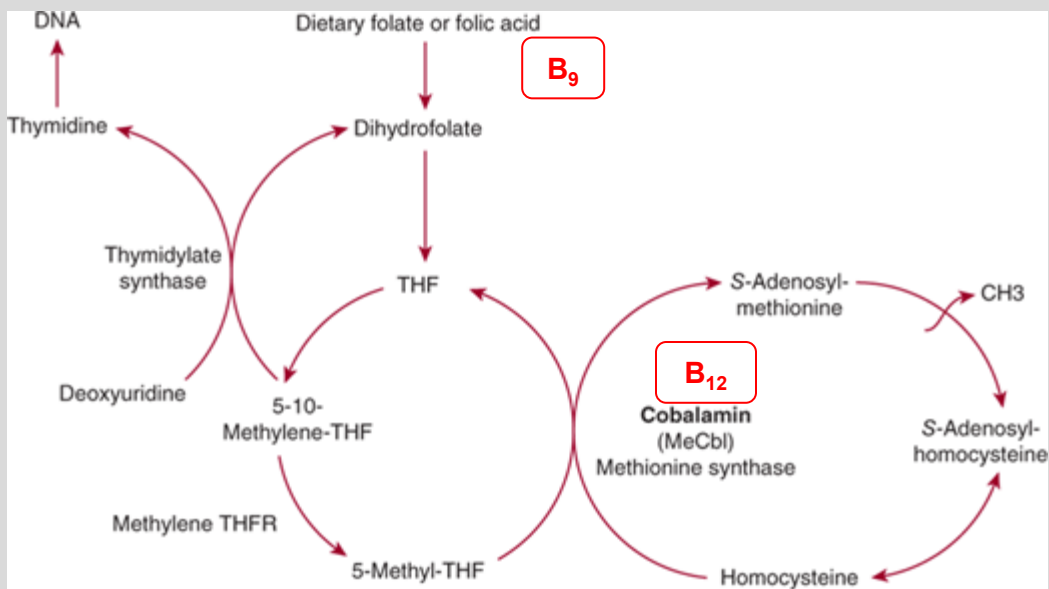
Megaloblastic Anemia Biochemistry

Takeaways:

- If **B₉** is deficient, there will be an **elevation** in **homocysteine**
 - Methyl-tetrahydrofolate is a reactant
- If **B₁₂** is deficient, there will be an **elevation** in **homocysteine**
 - B₁₂ helps the reaction from homocysteine to methionine

How do we tell the difference

- In **B₁₂** deficiency, there will ALSO be an increase in **methylmalonyl-CoA**
 - B₁₂ helps the reaction from methylmalonyl-CoA to Succinyl CoA



Deficiency	Homocysteine	Methylmalonyl-CoA
B ₉ (Folate)	↑	-
B ₁₂ (Cobalamin)	↑	↑

Macrocytic Anemias: Folate Deficiency



Folate Deficiency

- Dietary folate is obtained from green leafy vegetables and fruits
- Absorbed in the jejunum
 - There is a few months of storage of folate within the body

Causes of Deficiency:

- **Poor diet (the elderly & alcoholics)**
 - Alcohol will block the absorption of folate in the jejunum
- **Increased demand (pregnant women, cancer, hemolytic anemias)**
- **Drugs (methotrexate)**

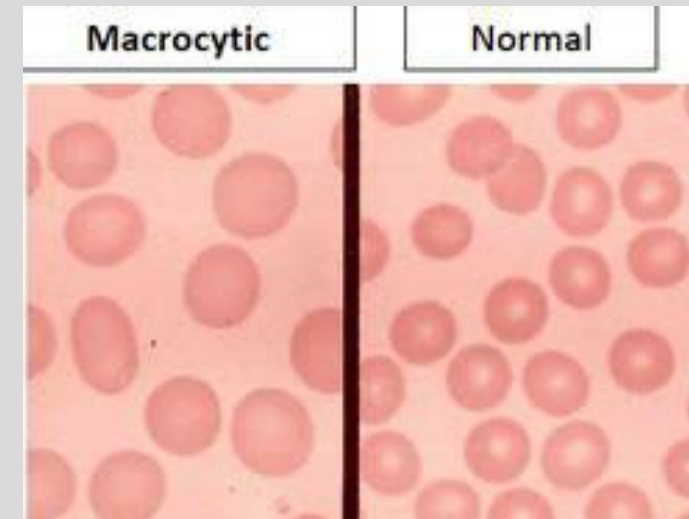
Clinical Findings:

- Macrocytic anemia and megaloblastic anemia
- Hypersegmented neutrophils
- Glossitis
- Folate levels: ↓
- Homocysteine ↑
- Methylmalonic acid: normal

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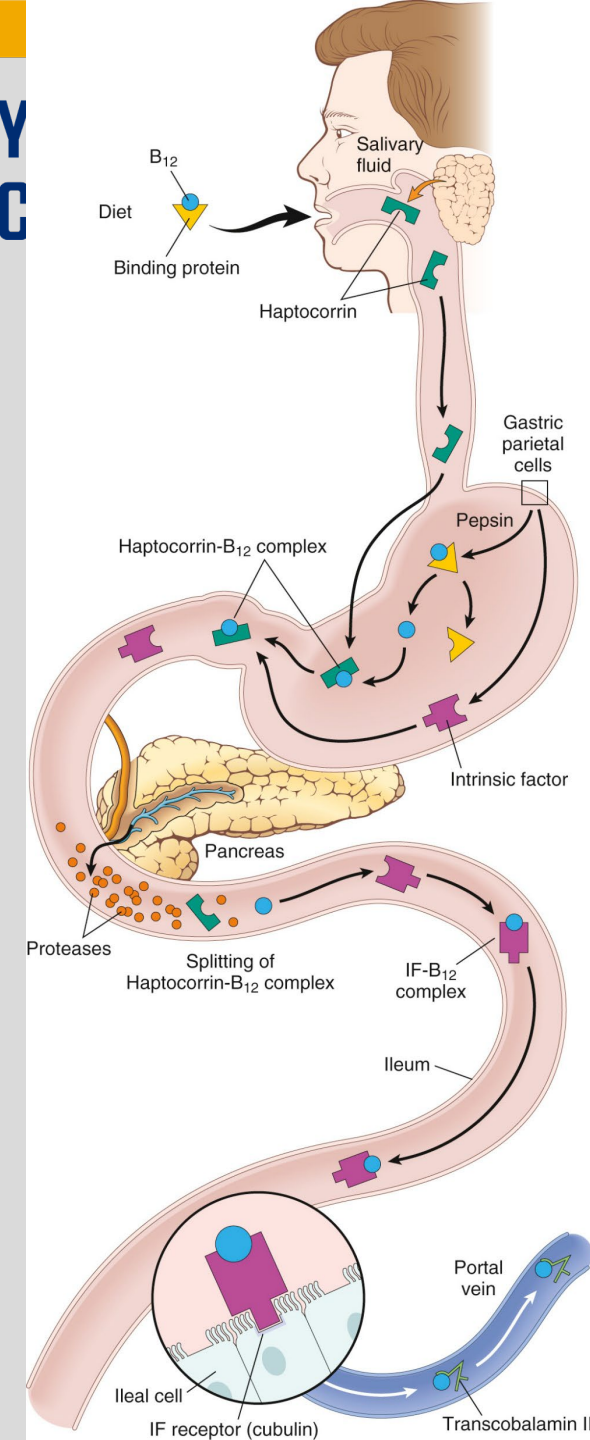


Source: Jon C. Aster, H. Franklin Bunn:
Pathophysiology of Blood Disorders, Second Edition
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Macrocytic Anemias: Vitamin B₁₂ Deficiency

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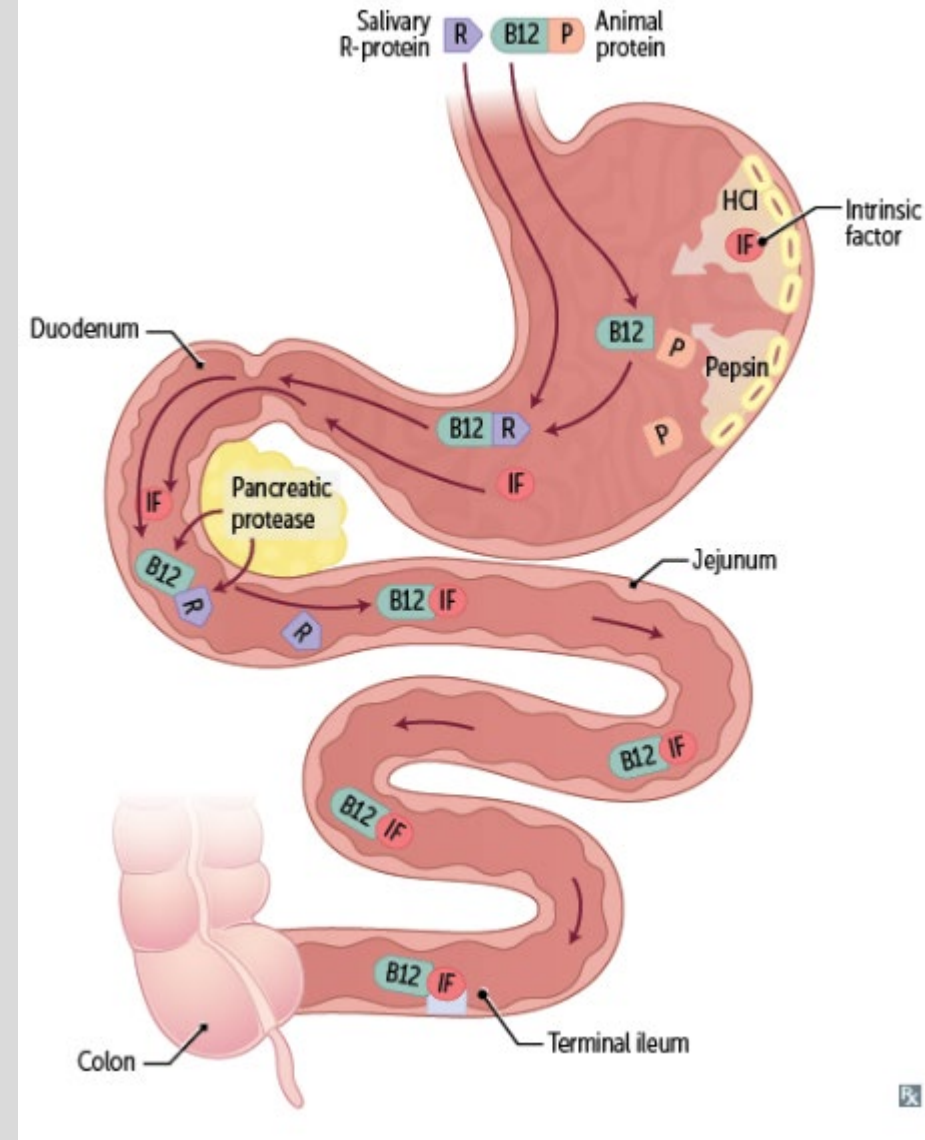


Dietary B₁₂ Absorption

- Dietary B₁₂ is complexed to **animal** derived proteins
 - This includes dairy and eggs
 - True vegans need B₁₂ supplementation

Pathway:

- B₁₂ travels to the stomach, where it is cleaved by pepsin (produced by chief cells as pepsinogen and activated by low pH)
- The salivary gland produces R-binder (haptocorrin), which proceeds to bind to B₁₂ in the stomach after it is cleaved
- **Parietal cells** produce **intrinsic factor**
- In the duodenum, pancreatic protease cleaves the B₁₂ from the R-binder protein, allowing **intrinsic factor** to bind to B₁₂
- This intrinsic factor-B₁₂ complex travels to the **terminal ileum**, where it is absorbed



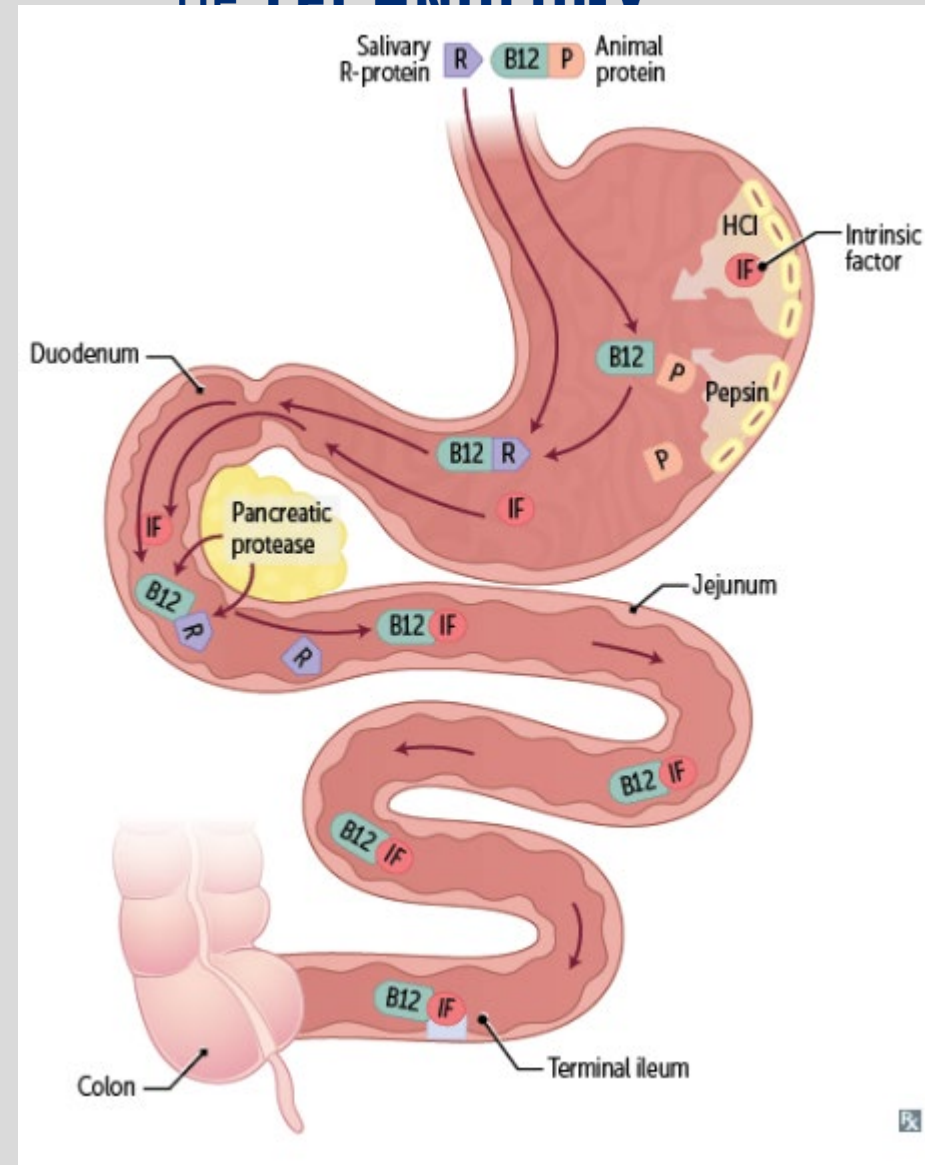


B₁₂ Deficiency

- B₁₂ deficiency is much less common than folate, as the liver has **years** of B₁₂ stored

Major Causes:

- Pure Dietary Deficiency: patients much be pure vegans for multiple years, and deny any supplementation
- Pernicious Anemia: **most common cause** of B₁₂ deficiency
 - Due to autoimmune destruction of the parietal cells (*atrophic gastritis: attack of autoantibodies on parietal cells*)
 - No parietal cells, **no intrinsic factor** and **increased stomach pH** → B₁₂ is not released from animal proteins
- Pancreatic Insufficiency: no protease to cleave B₁₂ from R-binder (chronic alcohol consumption)
- Terminal Ileum Damage: Crohn's disease (autoimmune) or tapeworms (*Diphyllobothrium latum*)
- Others: bacterial overgrowth





B₁₂ Deficiency: Findings

Lab Findings:

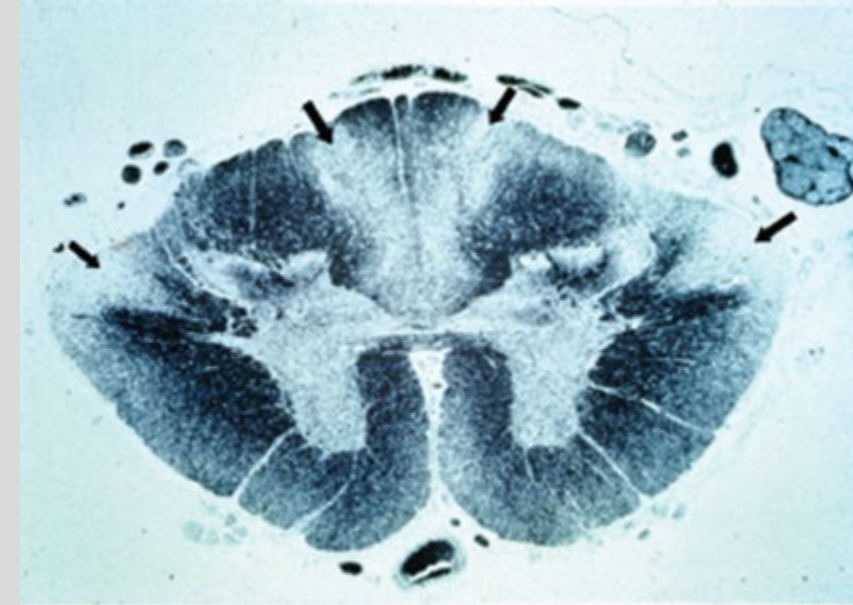
Macrocytic RBCs with hypersegmented neutrophils (just like folate deficiency)

- B₁₂ levels: ↓
- Homocysteine ↑
- Methylmalonic acid: ↑

Clinical Findings:

- Glossitis
- **Neurologic findings:** subacute combined degeneration
 - Poor proprioception and vibratory sensation (dorsal columns) and spastic paresis (lateral corticospinal tract)
 - Dementia
 - Due to **demyelination** of the posterior columns and lateral spinal tracts
 - B₁₂ is needed for the myelination of the central nervous system
- **B₁₂ deficiency anemia will respond to treatment with folate**
 - However, the **neurologic symptoms of B₁₂ deficiency will not resolve**, and may even be exacerbated

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Source: Jon C. Aster, H. Franklin Bunn:
Pathophysiology of Blood Disorders, Second Edition
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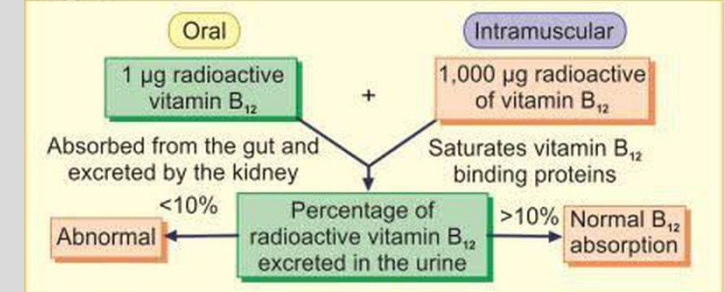
Determining Source (Historical)

Schilling Test:

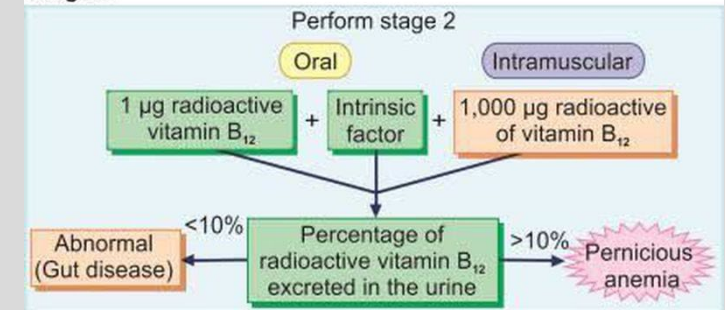
- **In the past**, part of the workup of B₁₂ deficiency
- Involves using orally consumed radiolabeled B₁₂ with and without intrinsic factor
 - Stage 1: radioactive B₁₂ without intrinsic factor
 - Stage 2: radioactive B₁₂ with intrinsic factor
 - Stage 3: radioactive B₁₂ without intrinsic factor after 7 days of Abx or given with pancreatic enzymes
- If B₁₂ is seen in the urine, we know the B₁₂ is able to be absorbed
- If B₁₂ is **not seen** in the urine, we know the B₁₂ is not able to be absorbed
- If the B₁₂ is only present when given with intrinsic factor, we know the issue is with intrinsic factor production (pernicious anemia)
- Test is not used much anymore as radioactive B₁₂ is used

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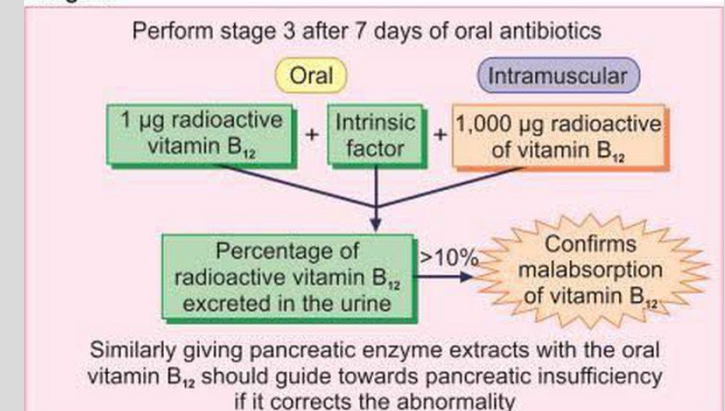
Stage 1



Stage 2



Stage 3



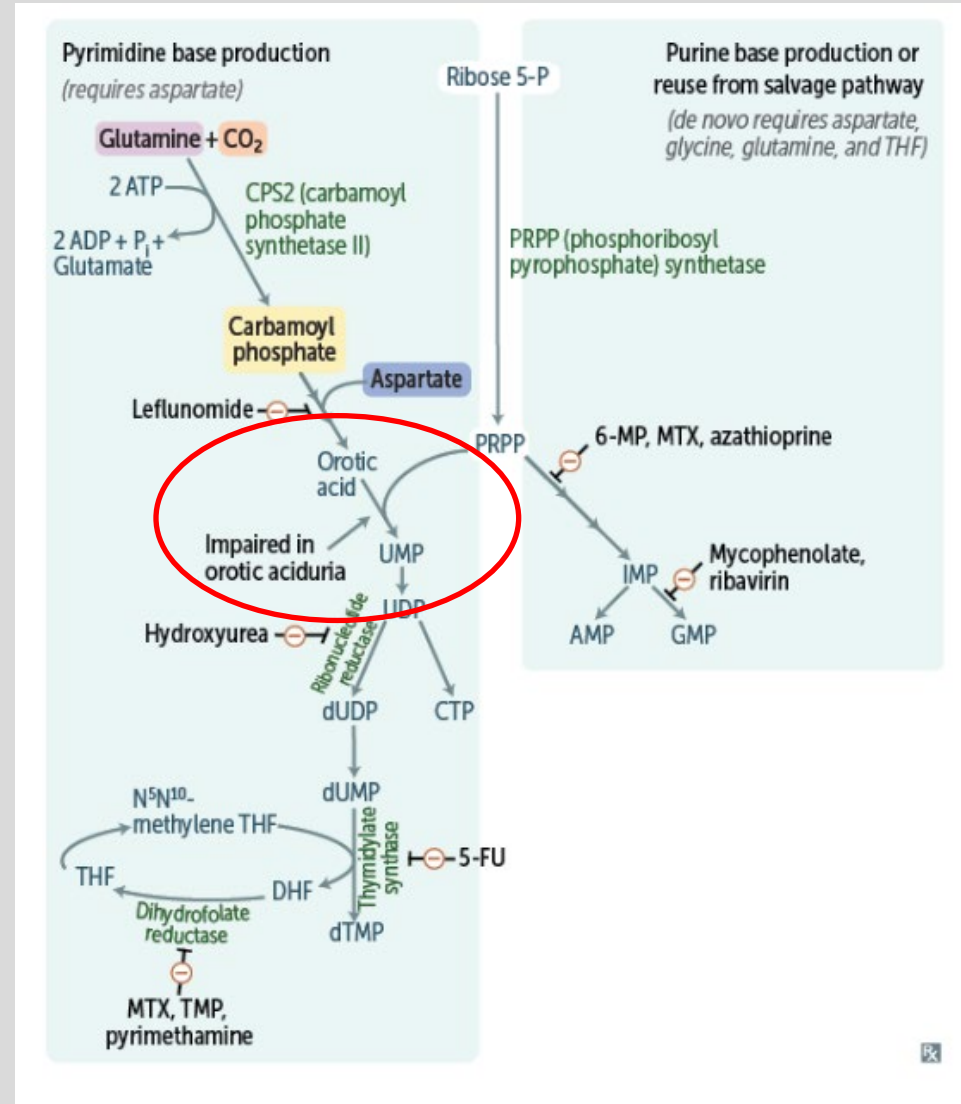
Macrocytic Anemias: Orotic Aciduria



Orotic Aciduria

- Inability to convert orotic acid to UMP due to defect in UMP synthase
 - Inability to produce pyrimidine → inefficient DNA synthesis
- Autosomal recessive
- Presentation
 - Failure to thrive and developmental delay
 - **Megaloblastic anemia** not treated by folate or B₁₂
 - This is still an issue with creating DNA
 - **Orotic acid in urine**
- **Treatment:** uridine monophosphate or uridine triacetate to bypass mutated enzymes

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Non-Megaloblastic Macrocytic Anemias: Diamond-Blackfan Anemia



Diamond-Blackfan Anemia

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- Congenital defect: **pure red blood cell aplasia**
- Intrinsic **defect** in **erythroid progenitor** cells leading to a decrease in erythroid progenitor cell numbers

Clinical Findings:

- Rapid onset anemia within the 1st year of life
- Short stature, craniofacial abnormalities, upper extremity abnormalities (triphalangeal thumbs)

Lab Findings:

- **↑** in HbF but **↓** total hemoglobin
- Bone marrow is devoid of RBC precursors



Other Causes

- Key difference: no issues with DNA synthesis
 - No hypersegmented neutrophils, but RBCs will still appear larger

Causes Include:

- Hemolysis or bleeding, causing reticulocytosis
 - The bone marrow is attempting to make more RBCs, pushing larger **immature reticulocytes out into circulation**
- Alcohol can damage the bone marrow
- Liver disease

Normocytic Anemias

(MCV 80-100)

Normocytic Anemia

- The RBCs are **normal** in size (MCV 80-100)
- Due to either:
 - Underproduction of RBCs
 - Increased peripheral destruction of RBCs
- We determine the cause via the **reticulocyte count**

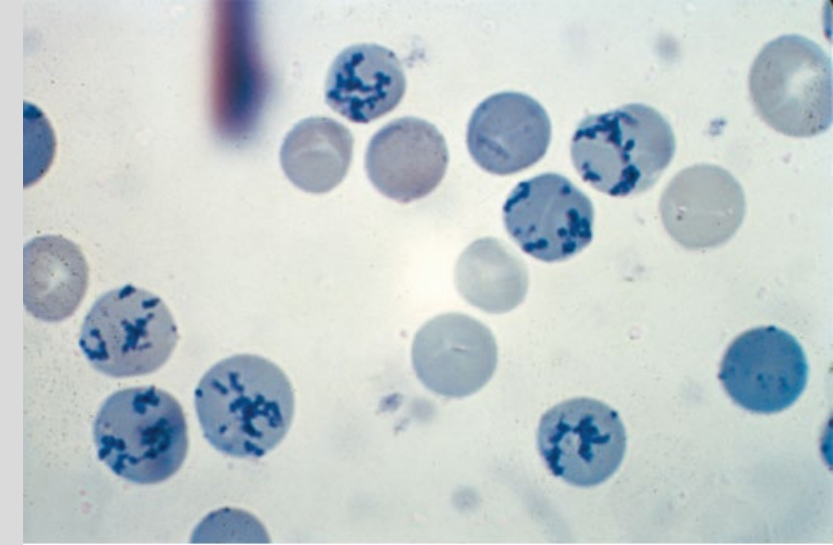
Reticulocytes:

- Young RBCs released from the bone marrow
- Appear on blood smear as larger cells with **blue cytoplasm**
- Normal reticulocyte count is ~1-2%

Reticulocyte Count:

- A properly functioning (normal) bone marrow will **increase** production, **increasing the reticulocyte count to >3%**
 - If the corrected reticulocyte count is <3%, we know the issue lies with RBC production
 - If the corrected reticulocyte count is >3, we know the issue is with peripheral destruction

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Source: Joseph Loscalzo, Anthony Fauci, Dennis Kasper, Stephen Hauser, Dan Longo, J. Larry Jameson: Harrison's Principles of Internal Medicine, 21e
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Normocytic Anemia Terms

Corrected Reticulocyte Count:

- Reticulocyte count is simple the % of total RBCs that are reticulocytes
- Anemia will FALSELY elevate the reticulocyte count
 - – **Reticulocyte / ↓ RBC → ↑ Reticulocyte Count**
- We correct this by multiplying the reticulocyte count by (**hematocrit/45**)
 - **Hematocrit:** measures the percentage of total blood volume that is RBCs

$$\frac{\text{Reticulocytes}}{\text{Total RBCs}} \times \frac{\text{hematocrit}}{45}$$

Hemolytic Anemias

Reticulocyte Count >3%



Hemolytic Anemia

Hemolysis Terms:

- **Intravascular:** breakdown of the RBCs occurs within the blood vessels
- **Extravascular:** breakdown of the RBCs occur within the reticuloendothelial system (macrophages of the spleen, liver, lymph nodes)
- **Intrinsic:** due to an issue with the RBC causing the destruction
 - ie: membrane/enzyme/hemoglobin issues
- **Extrinsic:** the RBC is fine
 - ie: aortic stenosis causing lysis of the RBC; autoimmune attack of normal RBC

Commonalities:

- Both intravascular and extravascular hemolytic anemias lead to:
 - ↑ lactate dehydrogenase: enzyme found RBCs
 - ↑ reticulocyte count
 - ↑ unconjugated bilirubin
 - Pigmented gallstones



Extravascular Hemolysis

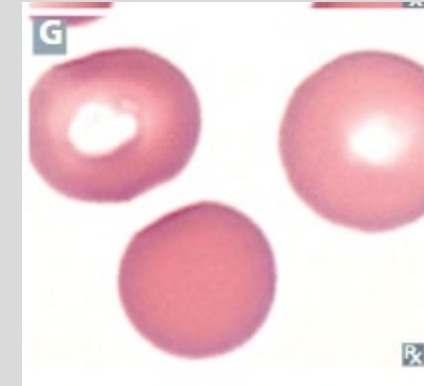
Extravascular Hemolysis

- Occurs in the reticuloendothelial system (macrophages of spleen, liver, lymph nodes)
- RBCs must pass through the spleen a few times per day, where they are inspected by macrophages
- Abnormal or senescent RBCs are consumed by macrophages
 - **Globin** is broken down to amino acids
 - **Heme** is broken down to iron and protoporphyrin
 - **Protoporphyrin** is broken down to unconjugated bilirubin → sent to liver attached to albumin for conjugation and excretion
- Abnormal Cells Removed:
 - Spherocytes and Sickle Cells
 - Howell-Jowell bodies
 - IgG/IgM attachment to the RBC

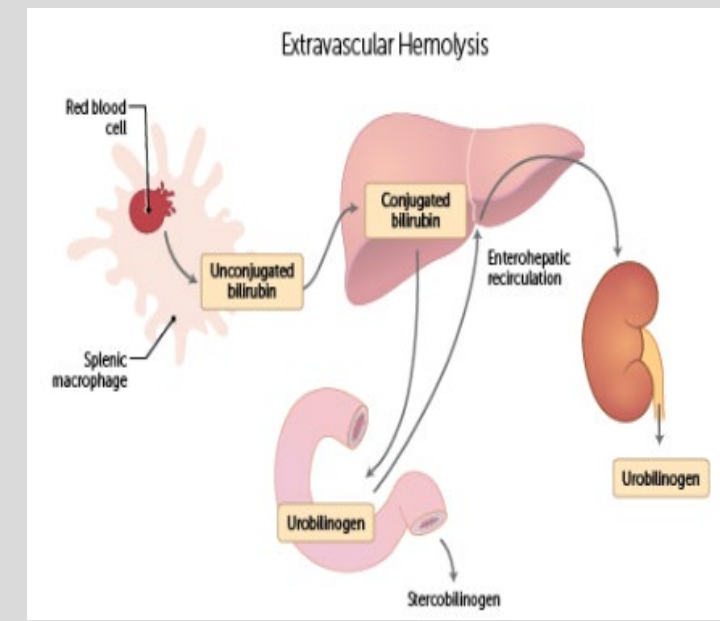
Clinical Findings:

- Spherocytes on peripheral blood smear
- Splenomegaly
- Elevate unconjugated bilirubin
- Jaundice and increased risk of gallstones
- **General cause:** usually due to alterations that make the red cells less deformable
 - Makes navigating the splenic sinusoids less successful, leading to red cell sequestration

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First Aid for the USMLE Step 1 2024





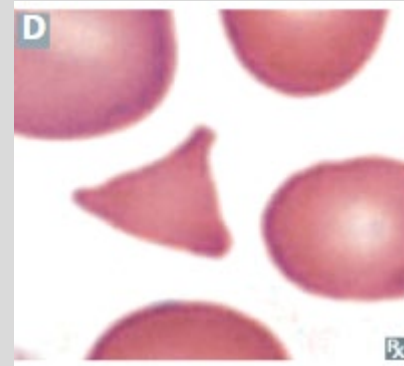
Intravascular Hemolysis

Intravascular Hemolysis

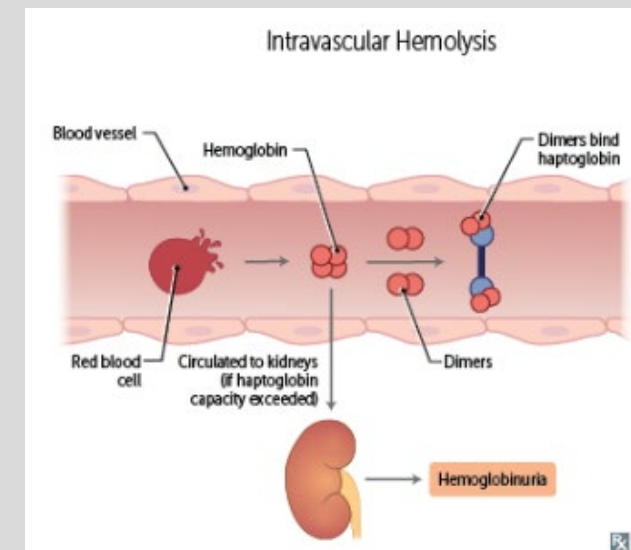
- Destruction within the blood vessel
- Free hemoglobin is circulating in the blood vessels → binds to **haptoglobin** for excretion

Clinical Findings:

- Schistocytes on peripheral blood smear
- Hemoglobinemia (free hemoglobin in blood)
- Hemoglobinuria (free hemoglobin in urine)
- Hemosiderinuria (hemosiderin (iron + protein) in urine)
 - Renal cells pick up some hemoglobin and break it down into iron
 - Iron binds to protein and is stored in the renal tubular cells
 - The tubular cells eventually shed, resulting in hemosiderinuria
- **Decreased** serum haptoglobin
 - Haptoglobin binds to free hemoglobin
 - Haptoglobin-heme complex is removed from circulation
- **General Cause:** mechanical injury, complement fixation, intracellular parasites, or exogenous toxic factors
 - **Mechanical Injury:** trauma by cardiac valves, narrowing of circulation due to thrombi, repetitive physical trauma (marathon runners)
 - **Completement:** IgM can activate complement



First Aid for the USMLE Step 1 2024



Intrinsic Hemolytic Anemias

Membrane Defect

Abnormal Hemoglobin

Deficient Enzymes

Hereditary Spherocytosis

Membrane Defect

Abnormal Hemoglobin

Deficient Enzymes

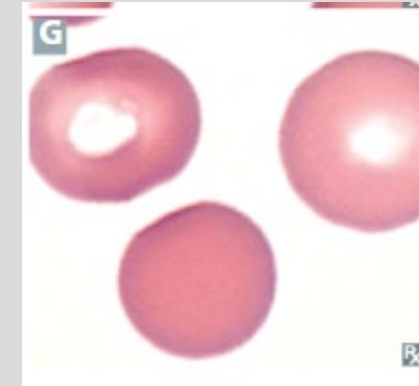
Hereditary Spherocytosis

Extravascular Predominance
Intrinsic Anemia

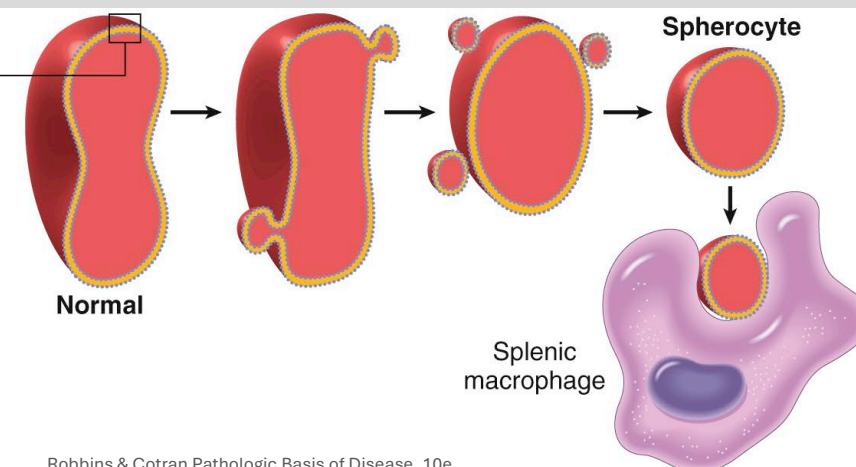
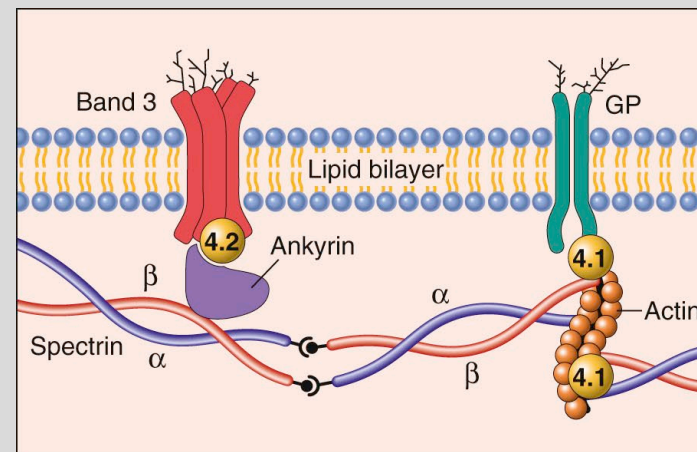
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Pathology:

- Autosomal dominant
- Defect in **proteins interacting with the RBC membrane** skeleton and plasma membrane
 - Most commonly ankyrin, spectrin, band 3, protein 4.2
 - Membrane is insufficient, and destabilizes the RBC
- The destabilized membrane creates blebs that are shed overtime, leading to decreased total membrane
 - Decreased total membrane removes the biconcave shape, making the cell round
(**SPHEROCYTE**)
- **Removed by the RES in the spleen (extravascular hemolysis)**, as spherocytes are less flexible to move through the splenic sinusoids



First Aid for the USMLE Step 1 2024



Robbins & Cotran Pathologic Basis of Disease, 10e

Hereditary Spherocytosis

Extravascular Predominance

Intrinsic Anemia

Clinical Findings:

- **Spherocyte** on blood smear (no central area of pallor!)
- **↑ RDW** and **↑** mean corpuscular hemoglobin concentration
- Splenomegaly, jaundice, and elevated unconjugated bilirubin (extravascular hemolysis)
- Gallbladder stones (due to increased bilirubin)

Complications:

- Increased risk for aplastic crisis with parvovirus B19

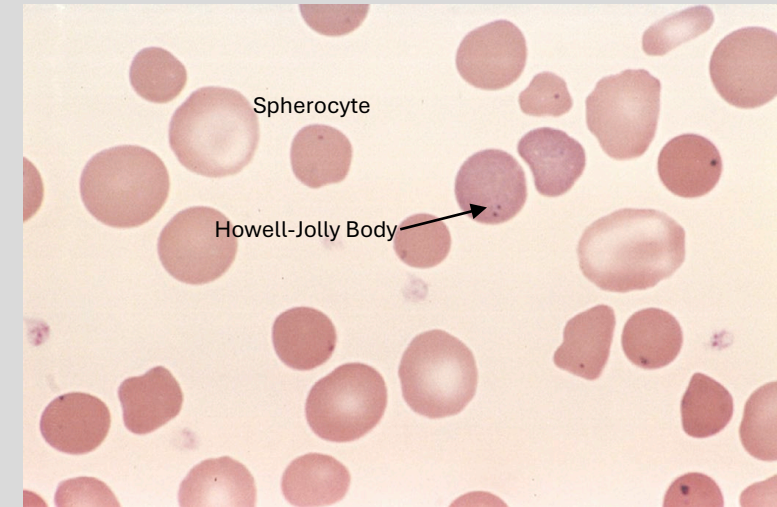
Diagnosis:

- **Osmotic fragility test:** place spherocyte in hypotonic solution. The RBC will lyse due to weakened membrane structure

Treatment:

- **Splenectomy** to resolve anemia
- **Spherocytes and Howell-Jolly bodies** (fragmented nuclear material in RBCs) will emerge on blood smear
 - Appear as small dark dots on PBS
- **Microbiology Correlation:** while splenectomy will resolve the anemia, it also places patients at **INCREASED** risk for infection by encapsulated pathogens (SHiN)
 - (**S**treptococcus pneumoniae; **H**emophilus influenzae, **N**eisseria meningitidis)

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Robbins & Cotran Pathologic Basis of Disease, 10e

Paroxysmal Nocturnal Hemoglobinuria

Membrane Defect

Abnormal Hemoglobin

Deficient Enzymes

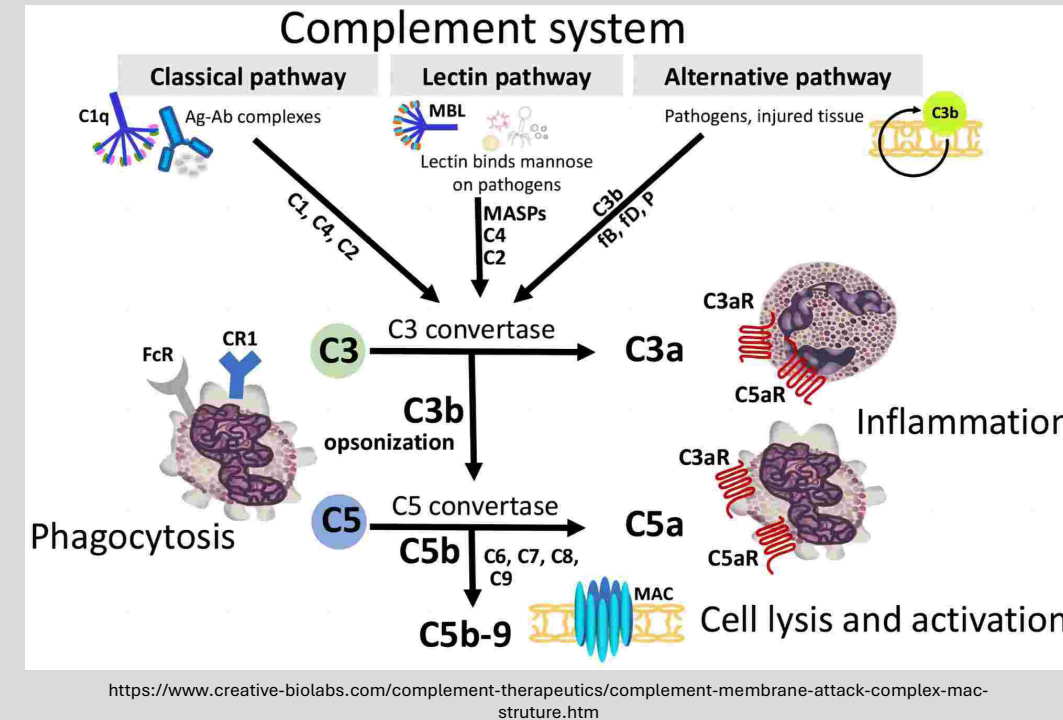
PNH

Intravascular Hemolysis

Intrinsic Anemia

Pathology:

- Defect in myeloid stem cell → **absent glycosylphosphatidylinositol (GPI)**, leaving the **cell susceptible to destruction by complement**
- GPI anchors decay accelerating factor on the surface of RBCs to protect them against complement
 - DAF inhibits C3 convertase**
 - Decay accelerating factor is also known as **CD55**
 - Absence of GPI leads to absence of DAF → complement can be activated
- GPI also anchors membrane inhibitor of reactive lysis (MIRL) to protect them against complement
 - MIRL is also known as **CD59**
- Typically occurs at night (*mild respiratory acidosis during sleep increases complement*)
 - DAF protects against complement activation
 - C3 convertase leads to the C5b-C9 membrane attack complex**
- Defect may affect other myeloid precursor cells (platelets, WBCs)



PNH

Intravascular Hemolysis
Intrinsic Anemia

Clinical Triad:

1. **Coombs Negative** hemolytic anemia
2. **Pancytopenia:** affects all myeloid precursors (neutropenia, thrombocytopenia)
3. **Venous thrombosis:** due to platelet destruction

Other Findings:

- Hemoglobinuria & hemoglobinemia (intravascular hemolysis)
- Hemosiderosis few days after hemolysis (intravascular hemolysis)
- **Pink/red** urine in the morning

Complications:

- Main cause of death is typically from thrombosis of hepatic, portal, or cerebral veins
 - Platelets release cytoplasmic contents into circulation
- Associated with aplastic anemia, acute leukemia

Diagnosis:

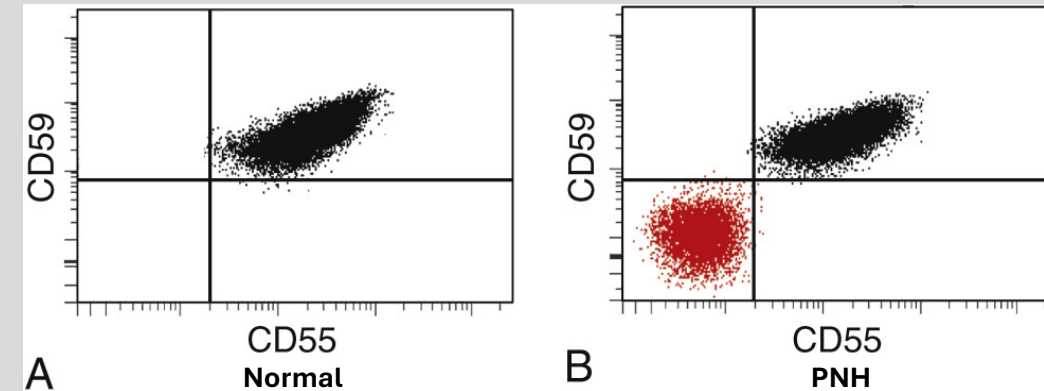
- **Flow cytometry** showing **absent CD55 & CD59**

Treatment:

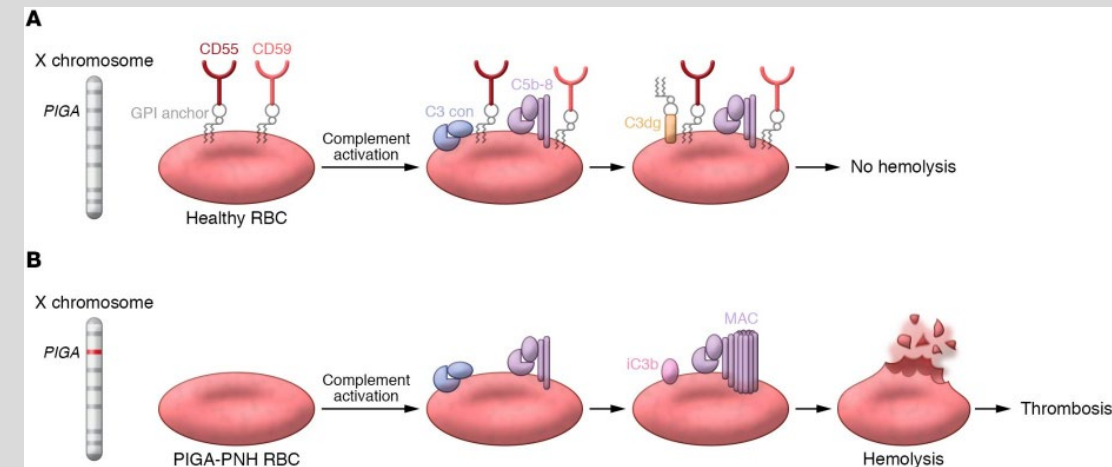
- Eculizumab (targets terminal complement protein C5)
 - Stopping membrane attack complex from forming further down the reaction pathway

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Robbins & Cotran Pathologic Basis of Disease, 10e



The Blood Project



Sickle Cell Anemia

Membrane Defect

Abnormal Hemoglobin

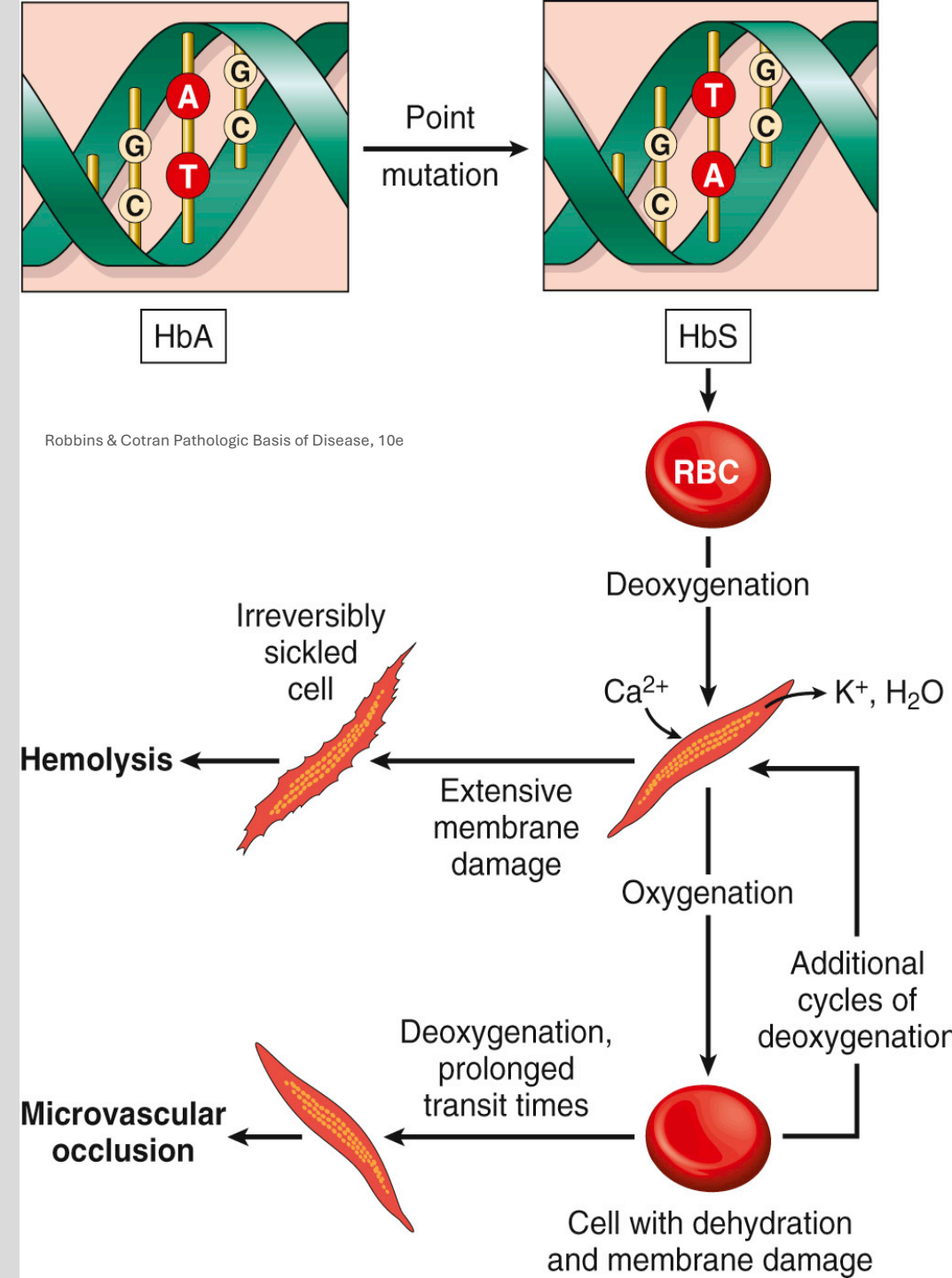
Deficient Enzymes

Sickle Cell Disease

Extravascular Hemolysis w/ some Intravascular
Intrinsic Anemia

Pathology:

- Autosomal recessive mutation in the β -chain of hemoglobin
 - Amino acid substitution: **glutamic acid** (hydrophilic) is replaced by **valine** (hydrophobic) due to a DNA point mutation
 - Abnormal folding causes aggregation of hemoglobin
- Carrier trait is protective for falciparum malaria
 - 10% of all individuals of African descent are carriers
- Disease arises with **two** abnormal β -globin genes are present
 - Results in >90% HbS hemoglobin in RBCs
 - HbS is still comprised of $\alpha_2\beta_2$ but β chains do not fold correctly
- When **deoxygenated**, the HbS molecules will stack into polymers, creating needle-like fibers
 - This stacking creates the classic “**sickle**” shape
 - Increased sickling will be seen with hypoxemia, dehydration, and acidosis
- Extensive damage will irreversibly sickle the cells, regardless of oxygenation state
- *Will not appear in newborn infants
 - HbF protects against sickling, as there is no β chain
 - High HbF protective for **first few months of life**
 - **First sickle cell crisis around ~6-9 months**



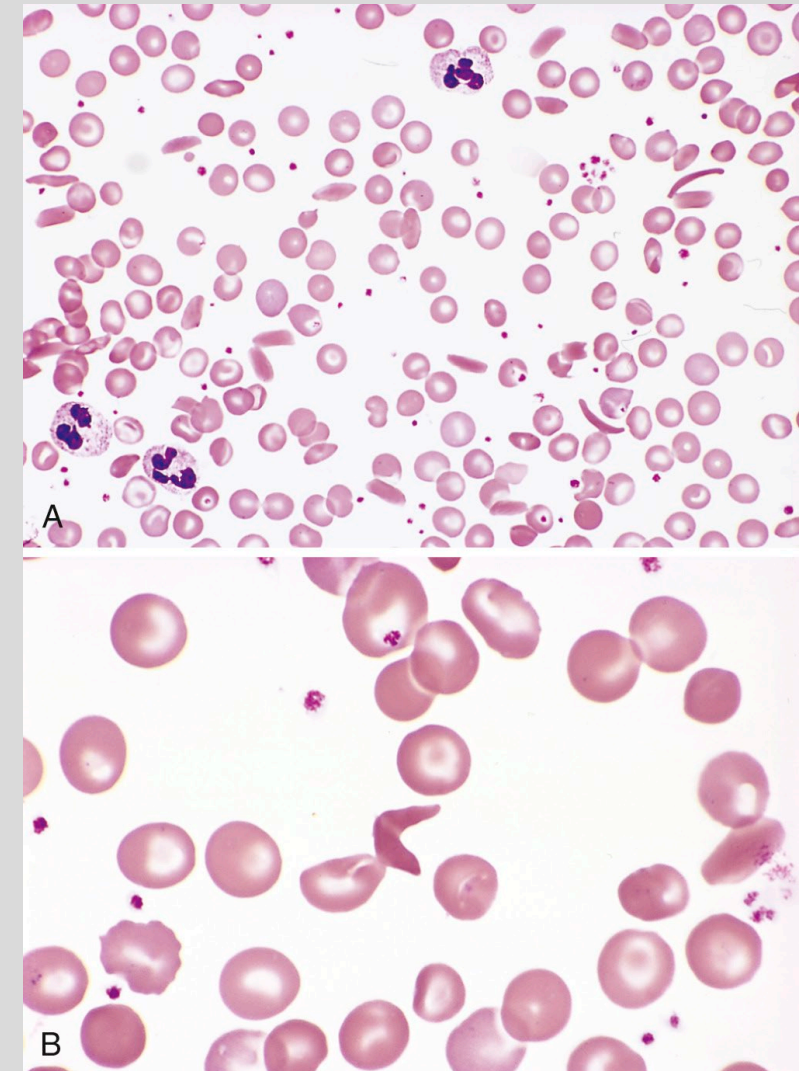
Sickle Cell Disease

Extravascular Hemolysis w/ some Intravascular
Intrinsic Anemia

Pathology:

- Overtime, **sickling will irreversibly damage the RBC membrane**
- **Extravascular hemolysis:** the reticuloendothelial system removes RBCs with damaged membranes and sickled cells
 - RBCs will become **sickled in hypoxic environments**, where the blood flow is slowed
 - Blood flow is **sluggish in the normal spleen and bone marrow**
 - **Anemia, Jaundice with unconjugated hyperbilirubinemia, Gallstones, Splenomegaly**
- **Intravascular hemolysis:** sickling leads to **membrane damage**, leaving the RBC mechanically **fragile** and susceptible to hemolysis
 - **Decreased haptoglobin, hemoglobinemia and hemoglobinuria**
- Bone marrow will be in **OVERDRIVE** to produce more RBCs: erythroid hyperplasia
 - Expansion of hematopoiesis into the skull (crewcut appearance on x-ray) and face (chipmunk facies)
 - Extramedullary hematopoiesis within the liver
 - Risk of aplastic crisis with infection of parvovirus B19
- **Key:** HbS will sickle in areas of low O₂

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Sickle Cell Disease

Extravascular Hemolysis w/ some Intravascular
Intrinsic Anemia

Complications:

- **Dactylitis:** swollen hands and feet due to vaso-occlusive infarcts in bones
 - Common sign in infants
- **Autosplenectomy:** spleen is enlarged in childhood due to red pulp congestion (trapping sickled RBCs in splenic cords and sinuses)
 - Chronic erythrostasis leads to splenic infarct, fibrosis, and shrinkage
 - **Howell-Jolly** bodies on blood smear (spleen usually removes these)
 - Similar to post splenectomy in hereditary spherocytosis
- Increased risk of **infection** (due to autosplenectomy)
 - **Encapsulated organisms:** SHiN (Strep pneumoniae, H. influenzae, N. meningitidis)
 - **Salmonella** paratyphi osteomyelitis
- **Acute Chest Syndrome:** vaso-occlusive crisis in pulmonary micro-circulation
 - Presents w/ chest pain, shortness of breath, lung infiltrates
 - Can be caused by pneumonia
 - *Most common cause of death in adults*
- Pain Crisis:
 - Vaso-occlusive crisis due to episodes of hypoxic injury and infarct/ischemia causing severe pain
 - Common sites: bones, lungs, liver, brain, spleen
- **Renal Papillary Necrosis:**
 - Renal medulla has lower PO_2
 - Hematuria (blood in urine)

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Robbins & Cotran Pathologic Basis of Disease, 10e



Sickle Cell Trait

- Sickle cell trait is the presence of one mutated and one normal β globin
 - The deciding factor in if the RBC will spontaneously sickle is the concentration of HbS in the cell; magic number is around 60%
- Hemoglobin:
 - HbA: 55%
 - HbS: 43%
 - HbA₂: 2%
- Generally asymptomatic: no anemia and no sickle cells in the GENERAL peripheral blood
- Sickling can still be present in the **renal medulla**
 - Renal medulla is extremely hypoxic and hypertonic
 - Leads to microinfarctions causing microscopic hematuria and decreased ability to concentrate urine



Sickle Cell Disease: Findings

Extravascular Hemolysis w/ some Intravascular
Intrinsic Anemia

Lab Findings:

- Sickle Cell Disease
 - Sickle cells and target cells
- Sickle Cell Disease and Trait
 - Metabisulfite screen: causes cells with any amount of HbS to sickle
- Hb electrophoresis
 - Disease:
 - HbS: 90%
 - HbF: 8%
 - HbA₂: 2%
 - Trait:
 - HbA: 55%
 - HbS: 43%
 - HbA₂: 2%

Treatment:

- Hydroxyurea; will increase HbF



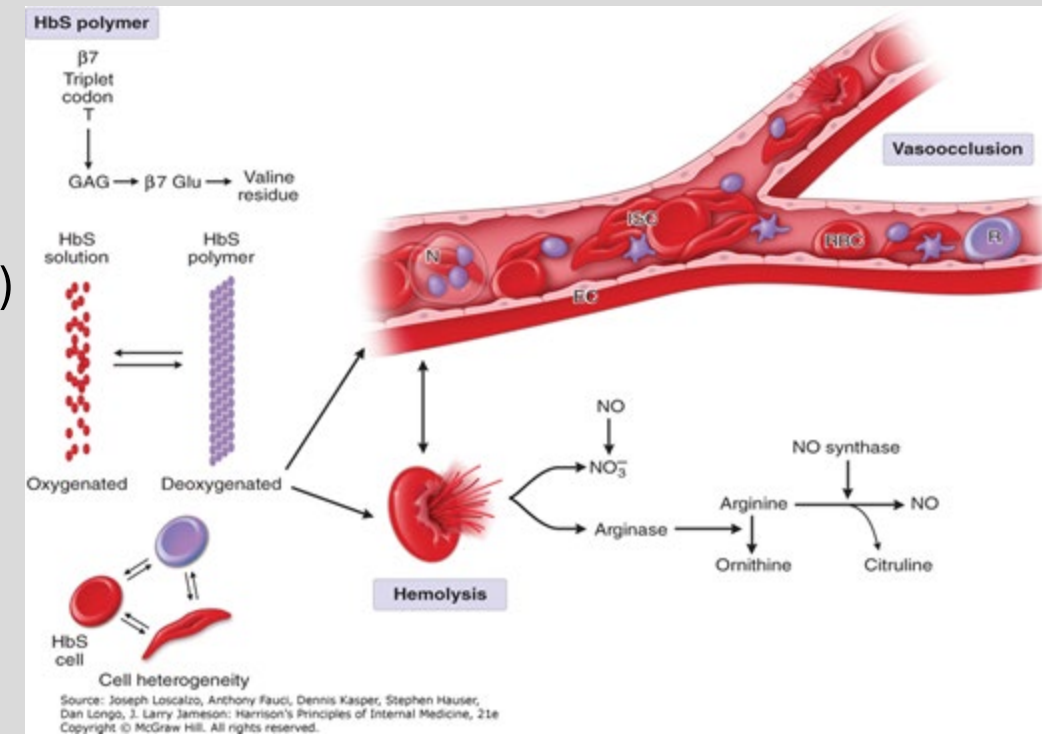
Sickle Cell Disease: Recap

Extravascular Hemolysis w/ some Intravascular
Intrinsic Anemia

Two Main Issues:

- **Severe Anemia:** RBCs are undergoing intrinsic extravascular hemolytic anemia
- **Vaso-Occlusive Disease:**
 - Sickle cells can block and obstruct microcirculation
 - Cause ischemia and pain to the area of obstruction
 - This is the cause of dactylitis in children and acute chest syndrome in adults
- Damage to the organs:
 - Kidney, spleen (autosplenectomy by early adulthood)
- Avascular necrosis

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Hemoglobin C Disease

Membrane Defect

Abnormal Hemoglobin

Deficient Enzymes

HbC Disease

Extravascular Hemolysis
Intrinsic Anemia

Pathology:

- Autosomal recessive
- B Globin Mutation: Glutamic acid to lysine mutation

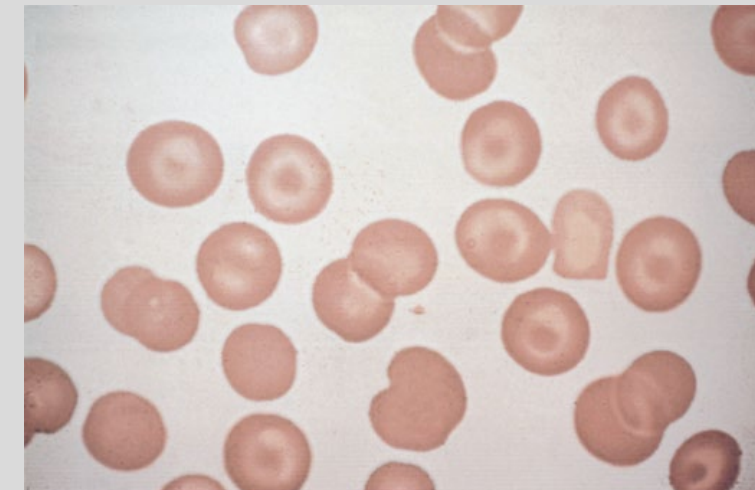
Presentation:

- Hemoglobin crystals inside RBCs
- Target cells

Hemoglobin Crystals:



Target Cells:



Source: Joseph Loscalzo, Anthony Fauci, Dennis Kasper, Stephen Hauser, Dan Longo, J. Larry Jameson: Harrison's Principles of Internal Medicine, 21e Copyright © McGraw Hill. All rights reserved.

Glucose-6-Phosphate Dehydrogenase Deficiency

Membrane Defect
Abnormal Hemoglobin
Deficient Enzymes

G6PD Deficiency

Intravascular Predominance w/ some Extravascular
Intrinsic Anemia

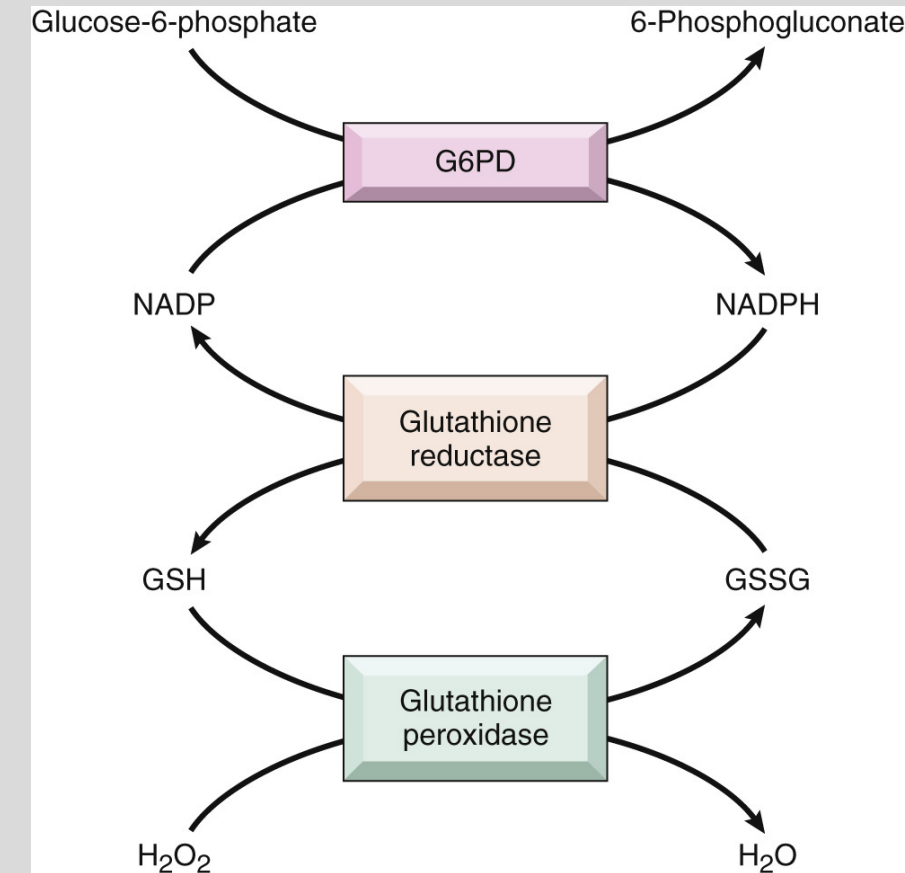
Pathology:

- X-Linked Recessive
- **Decreased half-life of G6PD**; renders RBC susceptible to **oxidative stress**
 - G6PD helps **produce NADPH, which will reduce glutathione**
 - **Glutathione** protects against oxidizing agents
- Hemolysis usually occurs during periods of **oxidative stress**
 - **Drugs**: primaquine, dapsone, sulfa drugs
 - **Food**: fava beans

Variants:

- **African Variant (mild)**: **mildly** reduced half-life of G6PD, leads to mild hemolysis with oxidative stress
 - Typically the older RBCs are susceptible, as they have less G6PD
- **Mediterranean Variant (severe)**: **markedly reduced** half-life of G6PD, leads to severe intravascular hemolysis with oxidative stress
- Genetic Advantage: carrier trait protects against falciparum malaria

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↓ G6PD → ↓ NADPH → ↓ reduced glutathione → Injury due to Oxidation → Intravascular Hemolysis

G6PD Deficiency

Intravascular Predominance w/ some Extravascular
Intrinsic Anemia

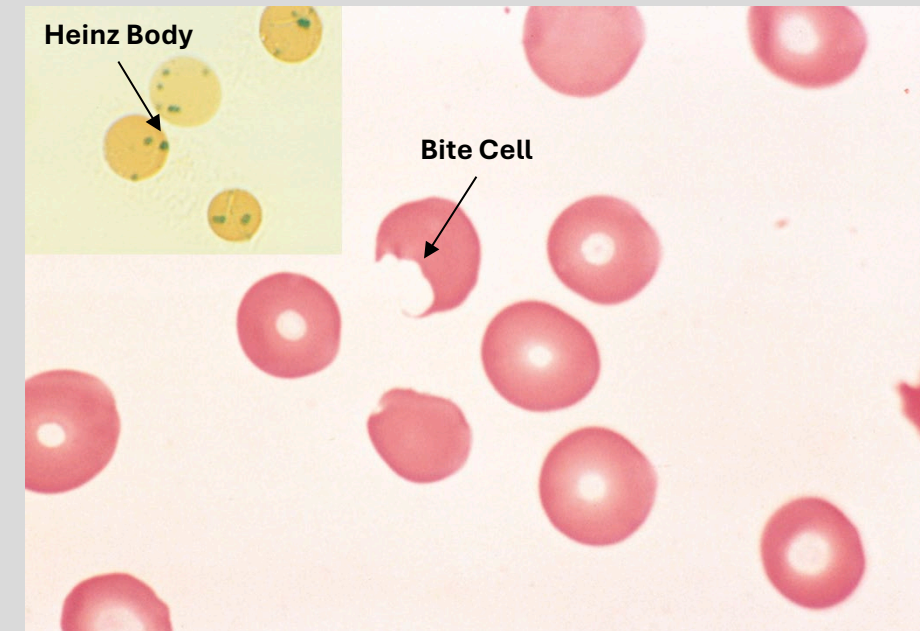
Clinical Findings:

- Peripheral blood smear **shows Heinz bodies and bite cells**
 - Heinz bodies: oxidative stress **precipitates hemoglobin**
 - Bite cells: **Heinz bodies are removed by spleen**, creating bite cells
- **Hemoglobinuria**
- Back pain (nephrotoxic)
- ↓ G6PD activity

Testing:

- Heinz preparation used to screen for disease
 - The hemoglobin precipitated as Heinz body can only be seen with special stain
- Weeks after hemolytic event: **enzyme testing**
 - G6PD activity may be falsely normal during acute hemolysis

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Pyruvate Kinase Deficiency

Membrane Defect
Abnormal Hemoglobin
Deficient Enzymes

Pyruvate Kinase Deficiency

Extravascular Hemolysis

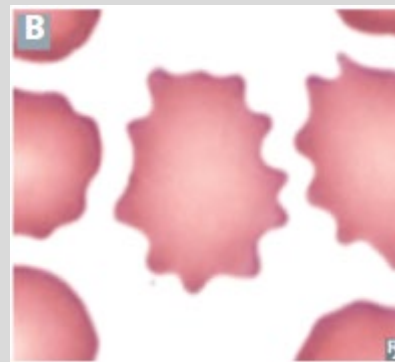
Intrinsic Anemia

Pathology:

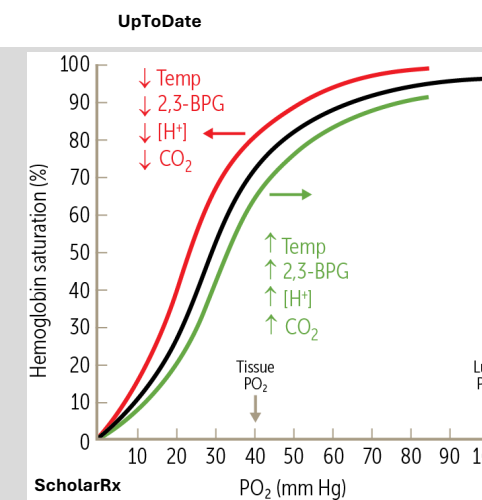
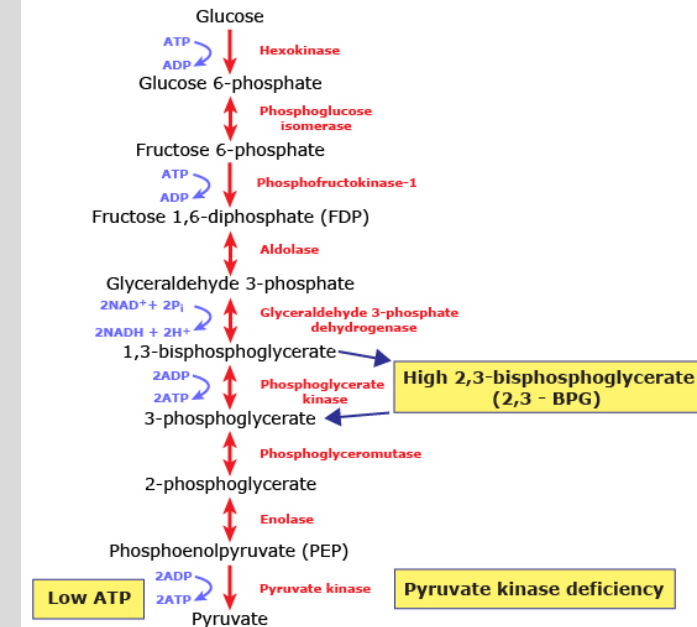
- **Autosomal recessive:** thoughts of PK deficiency being protective against falciparum malaria
- Decrease in pyruvate kinase → ↓ ATP → **rigid RBCs**
 - Rigid RBCs have a harder time navigating through the reticuloendothelial system
- Changes in **oxygen affinity**
 - Pyruvate kinase is also downstream of the 2,3 bisphosphoglycerate (BPG) shunt
 - By decreasing pyruvate kinase, there is a **buildup of BPG**
 - 2-3-BPG: increases oxygen delivery due to a rightward shift of the oxygen dissociation curve in hemoglobin

Clinical Findings:

- Hemolytic anemia, especially in newborn
- Peripheral blood smear: burr cells



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Extrinsic Hemolytic Anemias

Autoimmune Hemolytic Anemia



Autoimmune Hemolytic Anemia

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Extravascular Hemolysis

Extrinsic Anemia

Pathology:

- Antibodies that recognize red cells and lead to premature RBC destruction
- Broken down into warm and cold types

Diagnosis:

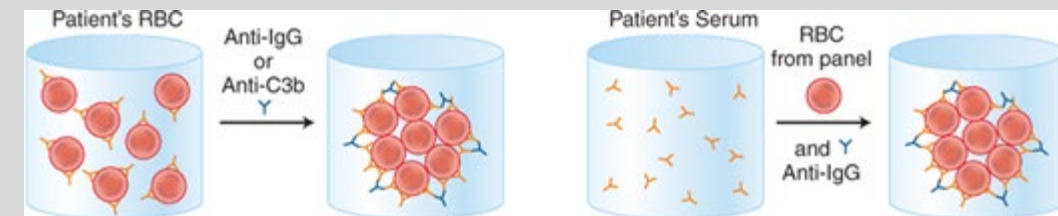
- Must perform a Coombs antiglobin test to detect antibodies and/or complement on red cells from the patient
 - Autoimmune Hemolytic Anemia is Coombs (+)

Coombs Test: detects the presence of antibodies against circulating RBCs

- Direct Antiglobin Test: **anti-human globin** (Coombs reagent) is **added** to the patient's RBCs.
 - RBCs Agglutination if RBCs are coated with anti-RBC antibody
 - Use: autoimmune hemolytic anemia
- Indirect Antiglobin Test: **normal RBCs added** to patient's serum with **Coombs agent**
 - RBCs Agglutination if serum has anti-RBC antibodies
 - Use: pretransfusion testing

Direct Coombs: are the RBCs coated with the Ab?

Indirect Coombs: are there Ab in the serum?



A. Direct

B. Indirect

Source: Jon C. Aster, H. Franklin Bunn:
Pathophysiology of Blood Disorders, Second Edition
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Autoimmune Hemolytic Anemia

Extravascular Hemolysis

Extrinsic Anemia

Warm Agglutination: mostly due to IgG

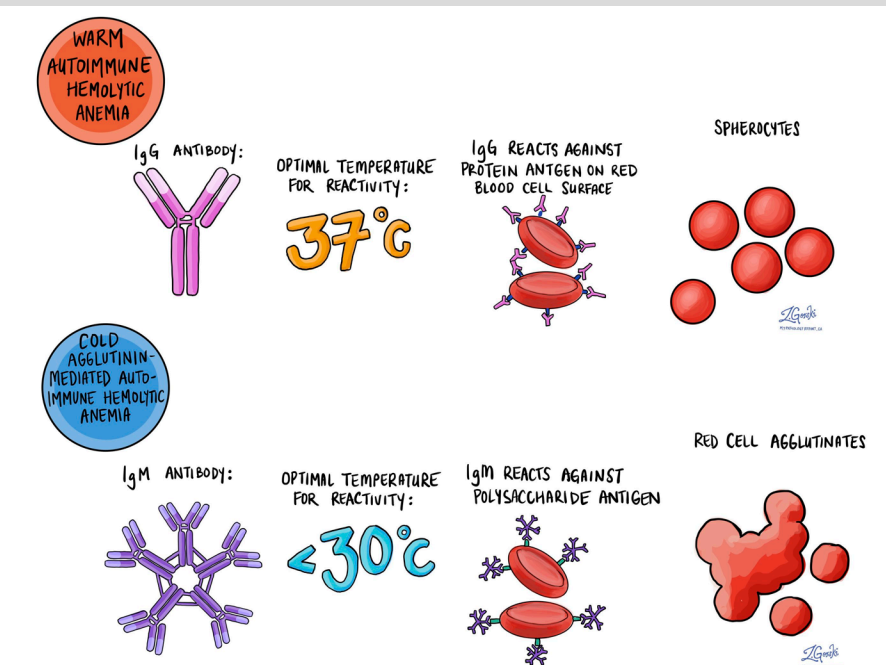
- IgG causes **extravascular hemolysis**
- Causes:
 - Idiopathic
 - Autoimmune disorders (Lupus)
 - Drugs (β -lactams, α -methyldopa)
 - Lymphoid neoplasms (CLL)
- Treatment: steroids, rituximab, splenectomy (refractory)

Cold Agglutination: mostly due to IgM

- IgM and complement cause RBC agglutination and extravascular hemolysis
 - RBCs can deactivate complement, but C3b acts as an opsonin in the spleen
- Seen with exposure to **cold** (especially in the limbs)
 - Painful blue fingers and toes caused by obstruction of agglutinated RBCs
- Causes:
 - **Mycoplasma pneumoniae**
 - Infectious **mononucleosis** (Epstein Barr Virus)
 - Chronic lymphocytic leukemia (CLL)
- Treatment: cold avoidance, rituximab

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Warm and Goopy cookies go with Cold Milk



Autoimmune Hemolytic Anemia

Extravascular Hemolysis

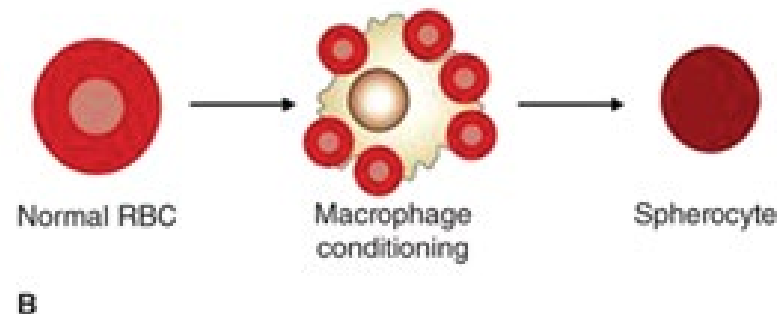
Extrinsic Anemia

Lab Findings:

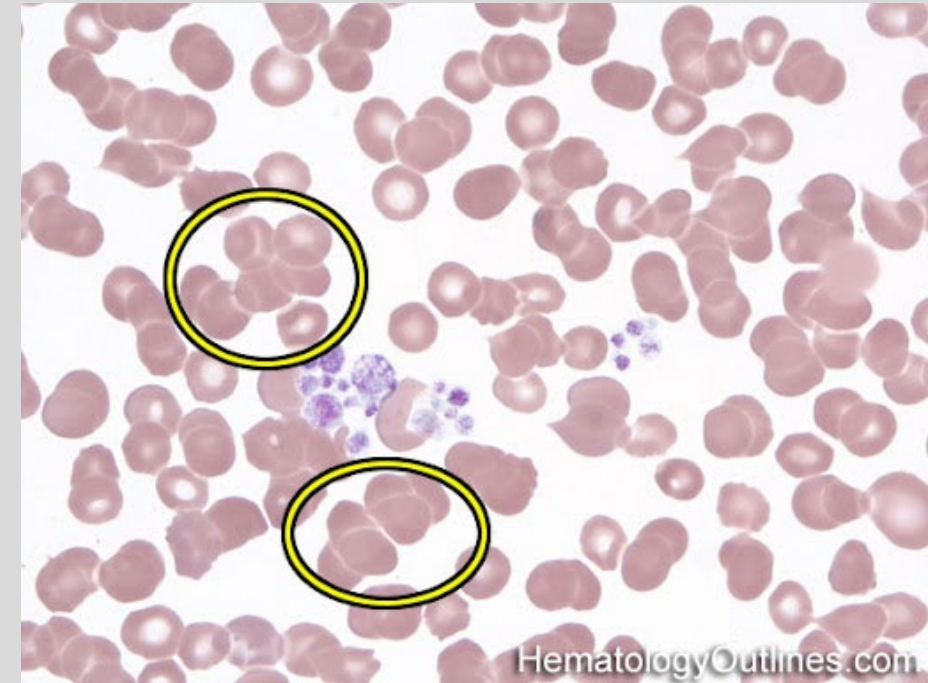
- **Spherocytes**
 - Similar action as heredity spherocytosis
 - Macrophages will **perform “partial” phagocytosis** of the RBC membrane
 - Decreased membrane → spherocyte
 - Spherocytes are sequestered and destroyed in the spleen
- **Agglutinated RBCs on peripheral blood smear**



Source: Jon C. Aster, H. Franklin Bunn:
Pathophysiology of Blood Disorders, Second Edition
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HematologyOutlines.com

Macro/Micro Angiopathic Hemolytic Anemia



Angiopathic Hemolytic Anemia

Intravascular Hemolysis

Extrinsic Anemia

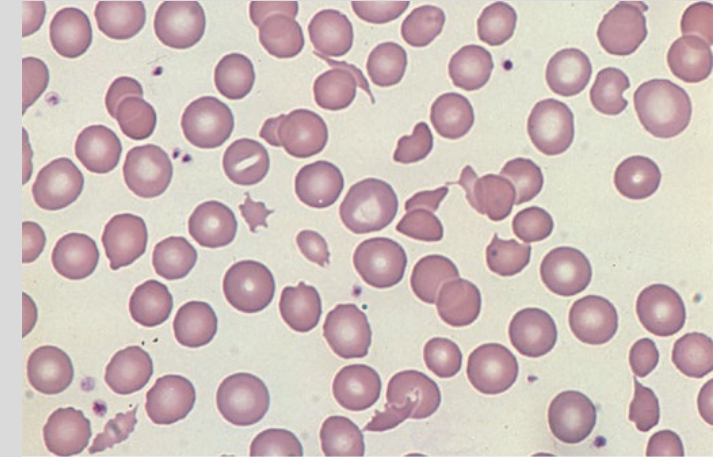
Macroangiopathic Hemolytic Anemia:

- Secondary to **mechanical destruction of the RBCs**
- Causes: prosthetic heart valve and aortic stenosis
- Peripheral blood smear: schistocytes
 - **Schistocytes**: caused by **mechanical destruction** of RBCs

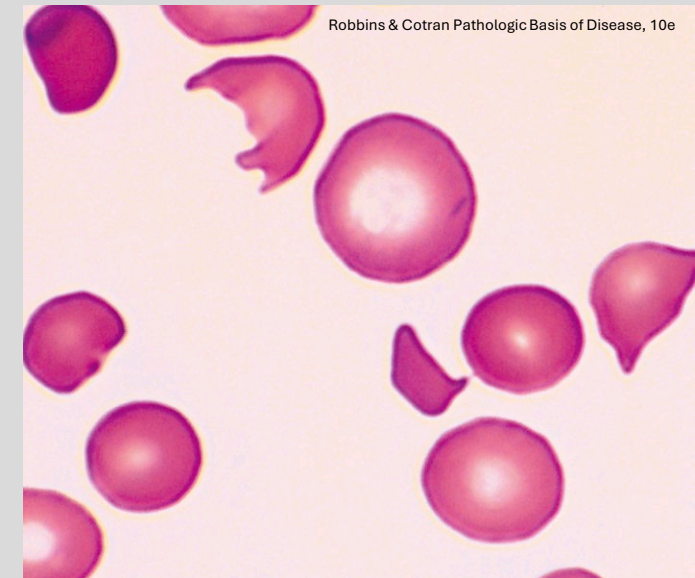
Microangiopathic Hemolytic Anemia:

- Secondary to **obstructed or narrowed vessels**
 - RBCs are damaged trying to squeeze through
- Causes:
 - Hypertensive emergency
 - DIC, TTP/HUS, SLE, HELLP
- Peripheral blood smear: schistocytes
 - **Schistocytes**: caused by **mechanical destruction** of RBCs

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Source: Joseph Loscalzo, Anthony Fauci, Dennis Kasper, Stephen Hauser, Dan Longo, J. Larry Jameson: Harrison's Principles of Internal Medicine, 21e
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Robbins & Cotran Pathologic Basis of Disease, 10e

Drug Induced Hemolytic Anemia



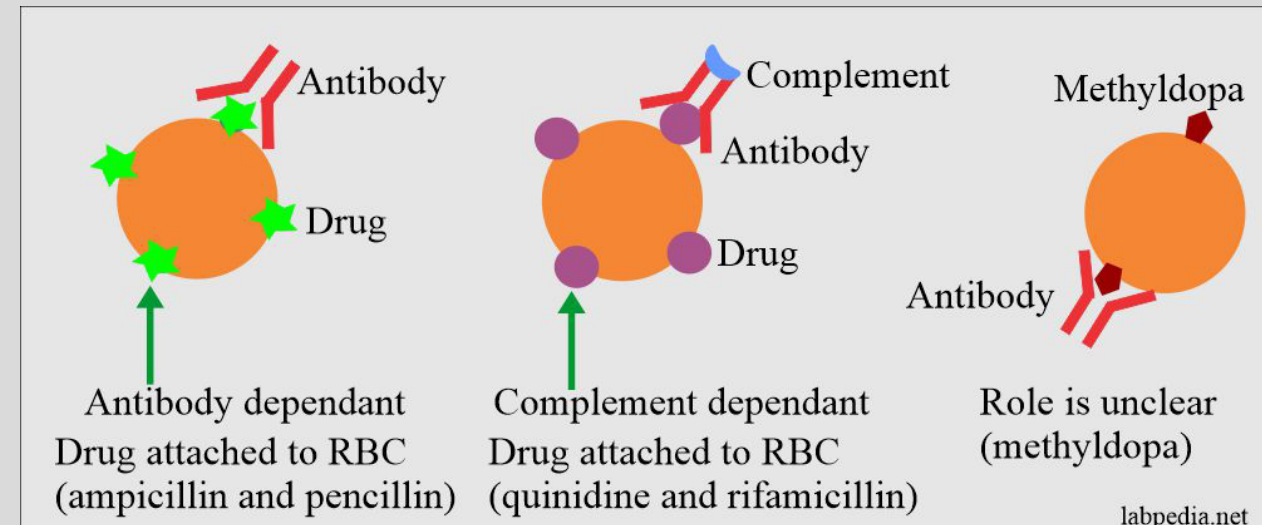
Drug Induced Hemolytic Anemia

Extravascular AND Intravascular Hemolysis Extrinsic Anemia

- Most commonly due to antibody-mediated immune destruction of the RBCs
- Can be due to oxidative injury due to free radicals
- Common causes:
 - Antibodies (penicillin, cephalosporins)
 - NSAIDs
 - Immunotherapy and chemotherapy

Lab Findings:

- Spherocytes: typically represent immune mediated destruction
 - Body treats the drug as a foreign substance, and attacks w/ antibodies
- Bite Cells: typically represent oxidative hemolysis (remove Heinz body)



Infectious Causes



Malaria/Babesiosis

Intravascular Hemolysis

Extrinsic Anemia

- Both cause **intravascular hemolysis**

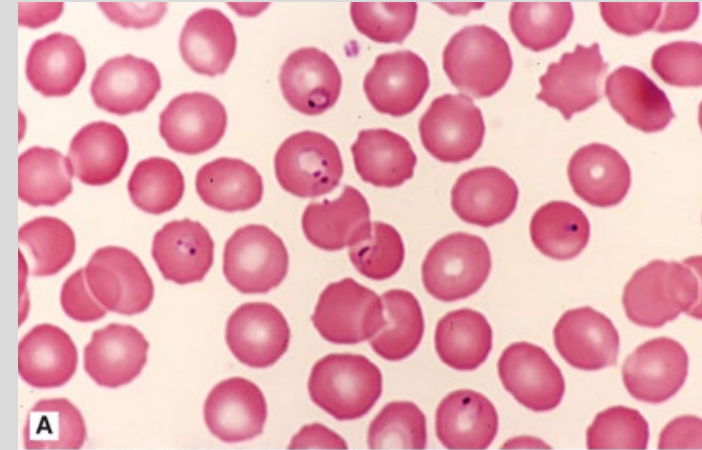
Malaria:

- Infection of RBCs and liver with Plasmodium
 - Transmitted by female *Anopheles* mosquito
- RBCs rupture as part of the plasmodium life cycle, causing intravascular hemolysis
- Key words: **cyclic fever** (daily, every other day)

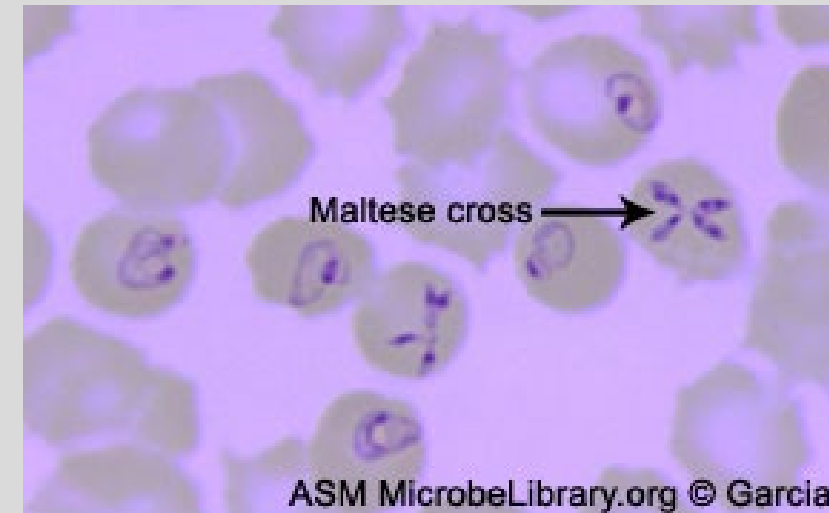
Babesiosis:

- Infection of RBCs
 - Transmitted by tick bite
- RBC will rupture, causing mild hemolytic anemia, fever, chills, malaise
- Higher risk of severe illness in asplenic patients
- Key words: Maltese cross in RBC, lives in northeast US

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Pathophysiology of Blood Disorders, Second Edition
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Anemia of Underproduction

Reticulocyte Count $<3\%$



Basic Principles

- Anemia is due to **decreased** RBC production
- Will see a decreased reticulocyte count

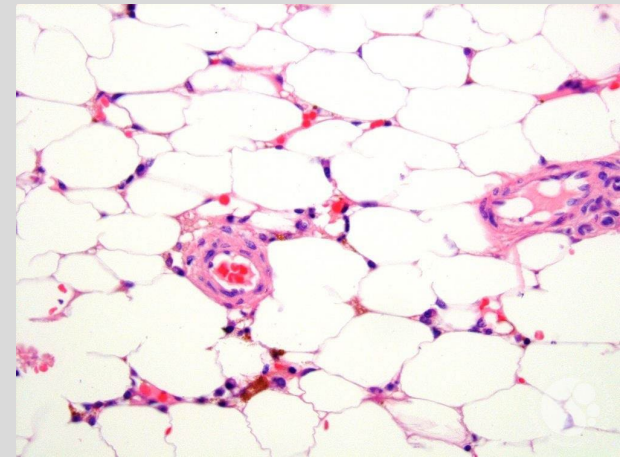
Common Etiologies:

- Causes of microcytic and macrocytic anemia (folate/iron deficiency)
- Renal failure
 - Peritubular interstitial cells in the kidney are responsible for EPO production
- Damage to bone marrow precursor cells

Specifics:

- Parvovirus B19: **infects progenitor RBCs** and **halts erythropoiesis**
 - No concerns for healthy patient, but if marrow is already under duress (**sickle cell, thalassemia**), can create aplastic crisis
 - Will present with slapped cheek rash
- Aplastic Anemia:
 - Hematopoietic stem cells are damaged, leading to pancytopenia
 - Anemia, neutropenia, thrombocytopenia w/ low reticulocyte count

ASH Image Bank





Aplastic Anemia

- Destruction/failure of the marrow

Causes:

- **Radiation**
- **Viral Agents** (EBV, HIV, Hepatitis)
- **Fanconi Anemia**
 - Autosomal recessive defect in DNA repair leading to marrow failure
 - Short stature, café-au-lait spots, thumb/radial defects, malignancy risk
- **Idiopathic**: immune mediated or stem cell defect
- **Drugs**: benzene, chloramphenicol, alkylating agents, antimetabolites

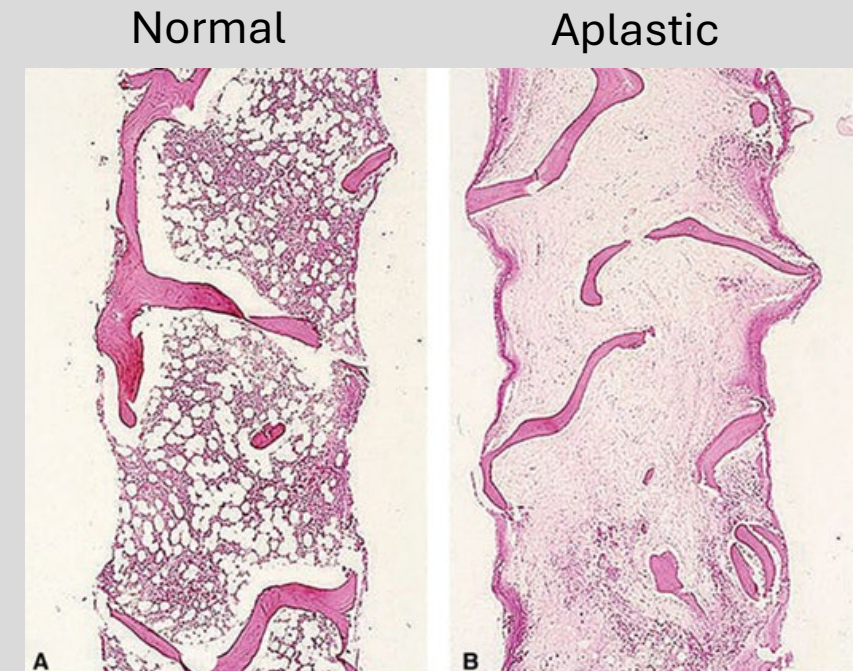
Lab Findings:

- Reticulocyte Count: ↓
- EPO: ↑
- Pancytopenia w/ normal cell morphology (shape)
- Bone Marrow: hypocellular marrow w/ fatty infiltration

Presentation:

- Fatigue, malaise, pallor, purpura, mucosal bleeding, petechiae, infection

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A B
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Pathophysiology of Blood Disorders, Second Edition
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RBC Characteristics

RBC Shapes

B: Echinocyte (Burr Cell)

- **Mechanism:** Pyruvate kinase deficiency
- **Pathology:** ATP depletion

C: Dacrococyte (Teardrop Cell)

- **Mechanism:** Bone marrow infiltration
- **Pathology:** Mechanical pressure as RBC leaves the marrow

D: Schistocytes (Helmet Cells)

- **Mechanism:** Angiopathic hemolytic anemia
- **Pathology:** Fragmented RBC due to mechanical stress

E: Bite Cells

- **Mechanism:** G6PD deficiency
- **Pathology:** Heinz body removal by the spleen

G: Spherocytes

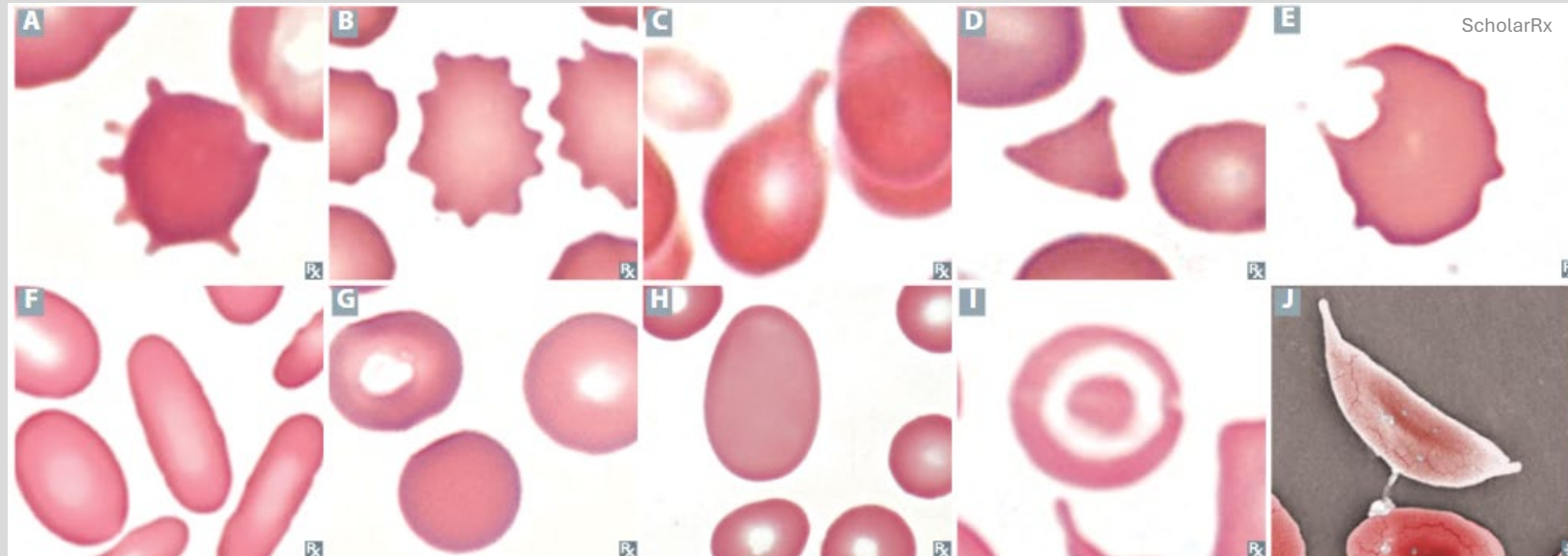
- **Mechanism:** Hereditary spherocytosis, autoimmune hemolytic anemia
- **Pathology:** ↓ surface area to volume ration: loss of membrane

I: Target Cells

- **Mechanism:** HbC Disease, Asplenia, Liver Disease, Thalassemia
- **Pathology:** ↑ surface area to volume: less hemoglobin/dehydration

J: Sickle Cell

- **Mechanism:** Sickle cell anemia
- **Pathology:** Misfolded β -globin



RBC Inclusions

Marrow:

A: Iron Granules:

- **Mechanism:** perinuclear mitochondria will contain excess iron
- **Pathology:** sideroblastic anemia
- **Notes:** need Prussian Blue stain to visualize

Peripheral Blood Smear:

B: Howell-Jolly Body:

- **Mechanism:** basophilic fragmented nuclear remnants
- **Pathology:** functional hyposplenism (usually removed by the spleen)

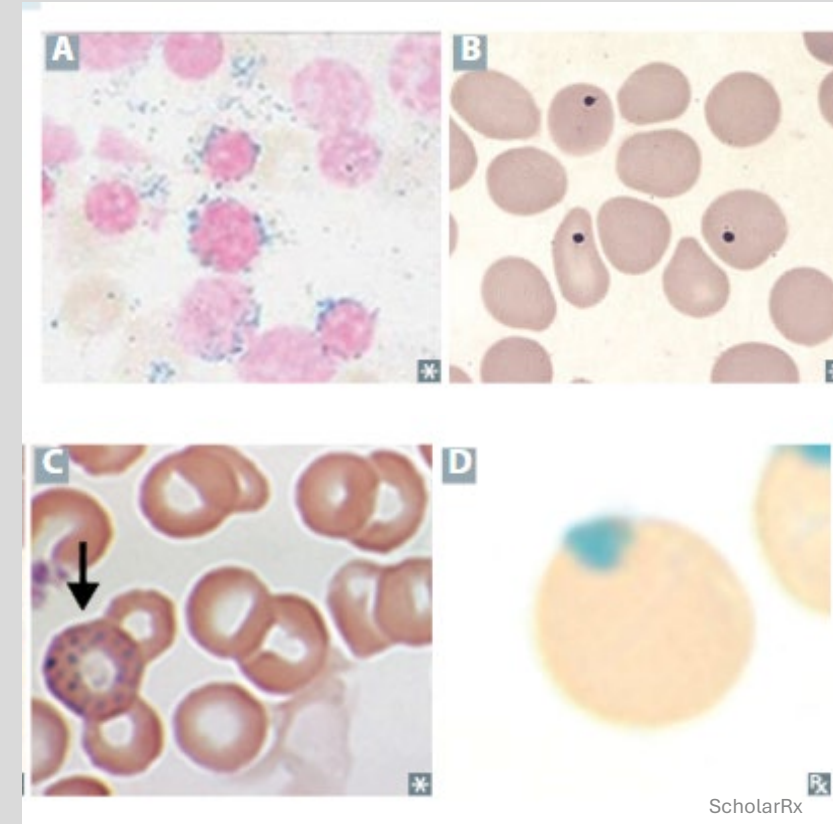
C: Basophilic Stippling:

- **Mechanism:** basophilic ribosomal precipitates (rRNA)
- **Pathology:** sideroblastic anemia; thalassemia

D: Heinz Body:

- **Mechanism:** denatured and precipitated hemoglobin due to oxidative stress, typically removed by the spleen (creates bite cells)
- **Pathology:** G6PD deficiency
- **Notes:** need supravital stain (crystal violet) to visualize

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Summary

Macrocytic Anemias (MCV >100):

- Most common megaloblastic (DNA issues): folate or B₁₂ deficiency
- Folate: (normal MMA & elevated homocysteine)
- Vit B₁₂: (elevated MMA & homocysteine), neurologic defects
- Blood Smear: hypersegmented neutrophils & larger RBCs

Normocytic Anemias (MCV 80-100 & Reticulocyte <3%)

- Anemia of Chronic Disease: ↑ hepcidin will sequester iron in marrow
- Aplastic Anemia: destruction of hematopoietic stem cells, causing pancytopenia

Normocytic Hemolytic Anemias (MCV 80-100 & Reticulocyte <3%)

Intrinsic Anemia:

- Hereditary Spherocytosis: defect in membrane proteins
 - Spherocytes; ↑ osmotic fragility
 - Extravascular hemolytic anemia (jaundice w/ unconjugated hyperbilirubinemia)
- Paroxysmal Nocturnal Hemoglobinuria: defect in CD55/59, leading to complement destruction
 - Coombs (–) hemolytic anemia, pancytopenia, thrombosis
 - Intravascular hemolytic anemia (↓ haptoglobin, hemoglobinemia and hemoglobinuria)
- G6PD Deficiency: sensitive to oxidative stress
 - Heinz body and bite cells
 - Extravascular w/ some intravascular hemolysis
- Pyruvate Kinase Deficiency: rigid RBCs
 - Lower hemoglobin affinity for O₂ w/ ↑ 2-3-BPG
 - Burr cells
- Sickle Cell Anemia: defect in β-globin chain (glutamic acid → valine)
 - Hypoxic, acidic, or dehydration will cause sickling → occlusion and anemia
 - Autospplenectomy, aplastic crisis, pain crisis, renal papillary necrosis
 - ↑ ↑ HbS w/ ↓ ↓ HbA
- HbC Disease: β-globin mutation (glutamic acid → lysine)
 - Blood Smear: hemoglobin crystals, target cells

Summary

Normocytic Hemolytic Anemias (MCV 80-100 & Reticulocyte <3%)

Extrinsic Anemia:

- Autoimmune Hemolytic Anemia
 - Warm: IgG causing extravascular hemolysis; caused by SLE, CLL, idiopathic
 - Cold: IgM + complement cause RBC agglutination and extravascular hemolysis w/ cold exposure
 - Raynaud's, CLL, Mycoplasma pneumoniae, Mononucleosis
- Drug-Induced Hemolytic Anemia
 - Usually Ab driven immune destruction, can be due to oxidative stress
 - Spherocytes (immune hemolysis) or Bite cells (oxidative stress)
 - Antibiotics, NSAIDs
- Microangiopathic Hemolytic Anemia: RBC damage via passage in obstructed or narrowed vessels
 - Schistocytes due to mechanical destruction
- Macroangiopathic Hemolytic Anemia: RBC damage via larger processes
 - Aortic stenosis, prosthetic heart valves
 - Schistocytes due to mechanical destruction
- Infection: malaria and *Babesia*

Question #1

A 24-year-old male presents with a history of fatigue, intermittent yellow skin, and left upper quadrant pain. Physical examination reveals jaundice and splenomegaly. The patient notes that his father was diagnosed with a type of hemolytic anemia. A peripheral smear reveals spherocytes with increased fragility when placed in a hypotonic solution. Which of the following is the most likely cause of this patient's condition?

- A. Autoimmune destruction of RBCs
- B. Defect in CD55 and CD59
- C. Defect in glucose 6-phosphate dehydrogenase
- D. Defect in spectrin
- E. Defect in the β -chain of hemoglobin

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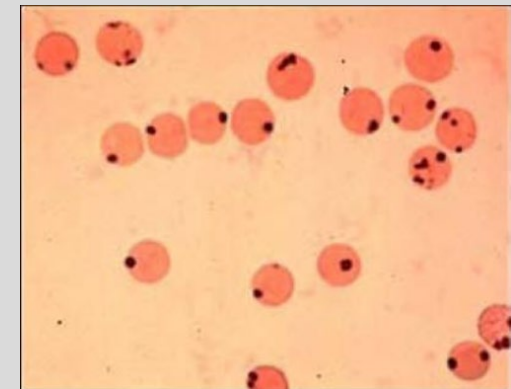
Question #2

A 14-year-old boy is brought to the clinic because of yellow discoloration of his eyes for 1 day. This developed 4 days after he had a 2-day illness marked by fever and watery diarrhea. Laboratory studies show:

Hemoglobin: 11 g/dL	(13.5-17.5 g/dL)
Hematocrit: 32%	(41-53%)
Platelet count: 250,000/mm ³	(150,000-400,000/mm ³)
Leukocyte count: 7000/mm ³	(4500-11000 /mm ³)
Reticulocyte count: 4.8%	(0.5-1.5%)
Indirect bilirubin: 3.6 mg/dL	(Total: 0.1-1.0 mg/dL)

A direct Coombs test is negative. A peripheral blood smear with crystal violet staining is shown. Which of the following is the most likely underlying mechanism of this patient's anemia:

- A. IgM and complement attaching to RBCs
- B. Decreased ability to reduce reactive oxidative species
- C. An inherited defect in ankyrin
- D. An inherited defect causing misfolding of β -chains
- E. Consumption of lead





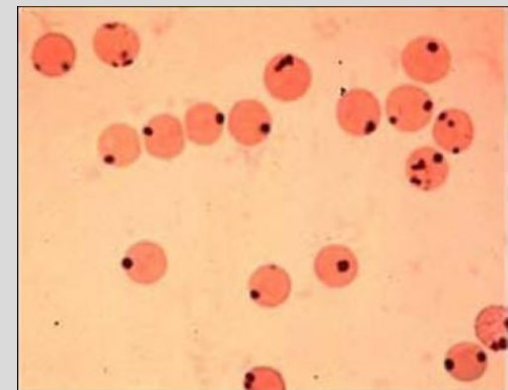
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- A. IgM and complement attaching to RBCs (spherocytes on PBS)
- B. Decreased ability to reduce reactive oxidative species (Heinz bodies)**
- C. An inherited defect in ankyrin (spherocytes on PBS)
- D. An inherited defect causing misfolding of β -chains (sickle cells)
- E. Consumption of lead (ringed sideroblasts)



Question #3

A 72-year-old woman comes to the clinic because of a 3-month history of fatigue. She lives alone and, over the past 6 months, has been consuming a diet of mostly tea, bread, and cheese due to financial concerns. She does not drink alcohol or take any medications. On physical examination, there is conjunctival pallor. The rest of the physical examination discloses no abnormalities. Results of complete blood count show:

Hemoglobin: 8.7 g/dL	(13.5-17.5 g/dL)
Hematocrit: 26%	(41-53%)
MCV: 113	(80-100)
Platelet count: 250,000/mm ³	(150,000-400,000/mm ³)
Leukocyte count: 7000/mm ³	(4500-11000 /mm ³)

Which of the following additional findings is most likely to be seen in this patient?

- A. Defects in proprioception and vibratory sensation
- B. Elevated serum levels of orotic acid
- C. Heinz bodies on peripheral blood smear
- D. Deep vein thrombosis
- E. Increased serum methylmalonic acid

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- C. Heinz bodies on peripheral blood smear
- D. Deep vein thrombosis**
- E. Increased serum methylmalonic acid

Question #4

A 65-year-old woman presents to her primary care physician reporting symptoms of generalized weakness, lightheadedness, and diarrhea for the past few months. She also reports a sensation of “pins and needles” in her feet. Medical history is significant for hypothyroidism. Physical examination reveals a decreased sensation to pinprick in her toes bilaterally and a positive Romberg sign. Pepsin levels are normal.

This patient is most likely deficient in a substance produced by which of the following types of cells?

- A. Chief cells
- B. G cells
- C. I cells
- D. Mucous cells
- E. Parietal cells



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Question #5

A 19-year-old female comes to clinic with complaints of fever, chills, malaise, fatigue, and joint pain. Earlier this week, she had a bright red rash on her face, describing it almost as a “slapped cheek” type of appearance. She has been in and out of the hospital multiple times throughout her childhood for sickle cell pain crises, and is on a stable dose of hydroxyurea. She states this does not feel like another exacerbation of her sickle cell. CBC results are as follows:

Hemoglobin: 8.7 g/dL	(13.5-17.5 g/dL)
Hematocrit: 15%	(41-53%)
Reticulocyte: 6%	(0.5-1.5%)
MCV: 90	(80-100)
Platelet count: 100,000/mm ³	(150,000-400,000/mm ³)
Leukocyte count: 4000/mm ³	(4500-11000 /mm ³)

What is the acute event causing the patient’s acute presentation.

- A. Parvovirus B19
- B. Genetic misfolding of hemoglobin
- C. RBC cytoskeleton insufficiency
- D. Chronic disease
- E. Autoimmune hemolytic anemia

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(Corrected: $6\% \times \frac{15}{45} = 2\%$)

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Question #6

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A 19-year-old college student presents to her PCP with complaints of fatigue, malaise, and lymphadenopathy. She states her roommates have been presenting with similar symptoms for the past few weeks. Physical exam is negative for any cardiovascular or respiratory abnormalities, but splenomegaly is seen on exam. The patient also states pain and paleness in her hands when walking outside in the winter without gloves. A CBC & PBS is performed w/ results below:

Hemoglobin: 9 g/dL	(13.5-17.5 g/dL)
Reticulocyte: 3.5%	(0.5-1.5%)
MCV: 90	(80-100)
RDW: Normal	
Leukocyte count: 15,000/mm ³	(4500-11000 /mm ³)

What is the pathophysiology of this patient:

- A. Autoimmune hemolytic anemia due to IgG antibodies
- B. Autoimmune hemolytic anemia due to IgM antibodies
- C. Underproduction of RBCs
- D. Chronic disease
- E. Membrane defect



Source: Jon C. Aster, H. Franklin Bunn:
Pathophysiology of Blood Disorders, Second Edition
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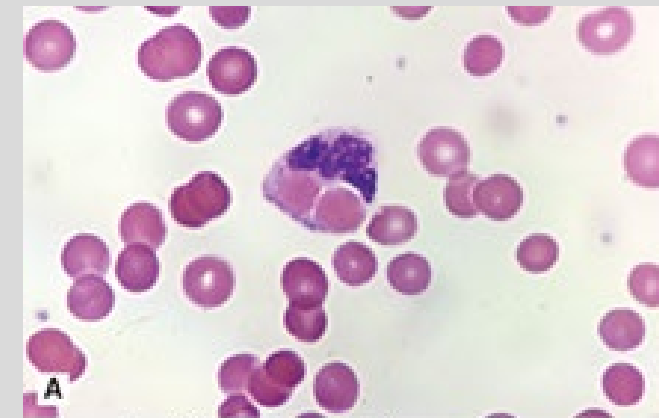
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